Exact Reconstruction and Higher Resonances in Polygamma–Zeta Calculus

Explicit even-zeta extraction, cubic tail moments,
Gamma determinants, and spectral logarithmic integrals

Research continuation prepared for the ProveIt project

11 October 2026 (UTC)

Abstract

This article continues the polylogarithm, harmonic-tail, Gamma, and Stieltjes programme at a fixed repository revision. Its main constructive result recovers every even-zeta coordinate from the optimal moment prefix for the pair of tail orders $(2, 2b)$, with no matrix inverse or unspecified minor left in the formula. The proof constructs an explicit Bernoulli reproducing kernel and evaluates its rank-one perturbation. The resulting polynomial weights give ordinary absolutely convergent polygamma series and finite polylogarithmic boundary identities. Further developments concern the optimal-prefix conjecture at tail order eight, all regular centered cubic trigamma moments, the next two-particle Gamma-determinant resonance, and spatial Mellin formulas for every spectral coefficient. At order eight an exact symbolic determinant certificate and a uniform Bernoulli congruence prove the conjecture outside one residue class modulo sixteen. The cubic moments close in four fixed constant coordinates. The resonance yields explicit generalized Stieltjes identities, shift derivatives, and an anchored Hurwitz-zeta primitive. All singular subtractions and coordinate conventions are stated explicitly. Exact finite calculations and independent numerical diagnostics accompany the proofs. The manuscript's short Gaussian S_6 and revised S_8 period identities remain unresolved; no numerical independence claim is made.

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1 Research target, source boundary, and conventions

1.1 What is continued here

The source for this investigation is the ProveIt manuscript and its six current incoming reports at commit

`446dca9bd65c20e409d1d2561d4ffad408103e2a`.

The source was inspected on 11 October 2026. The canonical manuscript already develops polylogarithmic identities, harmonic generating functions, Gamma correlations, and several finite-part calculi [1]. The incoming Bernoulli–Tornheim–digamma report proves optimal-prefix rank for the unequal even tail-order pairs $(2,q)$, $(4,q)$, and $(6,q)$, and asks for every reconstruction row in closed form [2, research question R2]. The first rational row is already known. This paper supplies the full inverse for $(2,q)$ and turns it into a family of evaluated ordinary series.

The six incoming archives were also inspected to distinguish new work from statements already supplied there. In particular, the general ordered-partition expansion of a product of tails, the symmetric cubic descent, the finite differential-polynomial integration-by-parts algorithm, and the first Gauss-determinant resonance are prior parts of this programme [3, 5, 4, 7, 6]. They are not relabelled as new discoveries in this article.

The word "new" below refers to the advancement beyond these inspected sources. A search of the relevant primary literature was used to identify classical ingredients; it does not establish global historical priority. The Bernoulli Hankel and orthogonal-polynomial background is classical [11, 12]. We provide the particular algebraic proofs needed here because the exceptional zeroth Bernoulli moment and the tail matrix's omitted binomial terms affect the answer.

1.2 Three different meanings of an evaluation

We distinguish an ordinary convergent sum, a meromorphically continued value, and a constant term of a specified cutoff. Where two of them coincide, that coincidence is proved. In the centered two-tail problem, the identities $\zeta(-2j,\frac{1}{2})=0$ make the requisite polynomial subtractions have zero continued sum. That special cancellation does not hold after an arbitrary change of centering.

The numerical checks in the accompanying files are diagnostics, not interval certificates. An identity in a polynomial ring over $\mathbb{Q}$ is checked exactly. A theorem about an infinite family or a continuum of parameters is justified by the written proof. No result in the paper uses a failed integer-relation search as an independence argument.

1.3 Notation

The Gamma function is Γ, the digamma function is $\psi=\Gamma'/\Gamma$, and $\psi^{(m)}$ is its mth derivative. For $p>1$ and $n\ge 0$, write

$$H_n^{(p)}=\sum_{j=1}^{n} j^{-p}, \qquad A_p(n)=\zeta(p)-H_n^{(p)}=\zeta(p,n+1), \qquad x_n=n+\tfrac{1}{2}. \tag{1.1}$$

For integer $p\ge 2$,

$$A_p(n)=\frac{(-1)^p}{(p-1)!}\psi^{(p-1)}(n+1). \tag{1.2}$$

The Hurwitz-zeta and polygamma conventions agree with [13, 14]. For $a > 0$, the generalized Stieltjes constants are normalized by

$$\zeta(1+u,a) = \frac{1}{u} + \sum_{j=0}^{\infty} \frac{(-1)^j}{j!}\gamma_j(a)u^j. \tag{1.3}$$

Thus $\gamma_0(a) = -\psi(a)$ and $\gamma_j = \gamma_j(1)$.

The Bernoulli convention is

$$\frac{te^{xt}}{e^t-1} = \sum_{j=0}^{\infty} B_j(x)\frac{t^j}{j!}, \qquad B_j = B_j(0), \qquad B_1 = -\tfrac{1}{2}. \tag{1.4}$$

The notation $(a)_j$ means a rising factorial. A falling factorial is $a^{\underline{j}} = a(a-1)\cdots(a-j+1)$. Empty products are one.

For $|z| < 1$, our depth-two polylogarithm is

$$\mathrm{Li}_{p,q}(z) = \sum_{n>m\geq 1} \frac{z^n}{n^p m^q}. \tag{1.5}$$

The color belongs to the larger index. This convention matters in the generating-function identities. At convergent endpoints we write $\zeta_{\mathrm{MZ}}(p,q) = \sum_{n>m\geq 1} n^{-p}m^{-q}$ for a multiple zeta value; this differs from the Hurwitz function $\zeta(s,a)$. The spectral double series $Z(s,t)$ in Section 8 uses the same strict ordering on its stated convergence domain. Auxiliary coefficients such as C_j and R_j are defined locally in each family and do not identify constants from different sections. All functions on the open unit disc are defined by their series; real boundary limits are taken with $z \uparrow 1$. Complex powers of positive real numbers use the real logarithm. Additional complex-parameter domains are stated locally where they are used.

2 The exact tail matrix and its regularization

Fix an integer $b \geq 2$. Consider the Dirichlet series

$$D_b(s) = \sum_{n=0}^{\infty} A_2(n)A_{2b}(n)x_n^{-s}, \qquad \mathrm{Re}\, s > 1-2b. \tag{2.1}$$

Its centered moments are denoted by $D_{b,r} = D_b(-2r)$, $r \geq 0$. At a point outside the displayed convergence half-plane, this notation means meromorphic continuation, not the original divergent series.

2.1 Why every even moment is regular

Define rational numbers

$$t_{p,0} = \frac{1}{p-1}, \qquad t_{p,j} = \frac{B_{2j}(1/2)(p)_{2j-1}}{(2j)!} \quad (j \geq 1), \qquad a_{b,j} = \sum_{i=0}^{j} t_{2,i}t_{2b,j-i}. \tag{2.2}$$

Euler–Maclaurin summation in the midpoint coordinate gives, to any fixed order and with its usual power remainder,

$$A_p(n) \sim \sum_{j\geq 0} t_{p,j}x_n^{1-p-2j}, \tag{2.3}$$

$$A_2(n)A_{2b}(n) \sim \sum_{j\geq 0} a_{b,j}x_n^{-2b-2j}. \tag{2.4}$$

Only even corrections occur, since $B_{2j+1}(1/2) = 0$. The first two product coefficients are

$$a_{b,0} = \frac{1}{2b-1}, \qquad a_{b,1} = -\frac{2b^2-b+1}{12(2b-1)}. \tag{2.5}$$

Proposition 2.1 (Ordinary subtraction at every even moment)**.** *The only possible poles of $D_b(s)$ are at $s = 1-2b-2j$, $j \geq 0$. In particular every $D_{b,r}$ is regular. For an even polynomial $W(x) = \sum_r w_r x^{2r}$, put*

$$R_{b,W}(x) = \sum_r w_r \sum_{j=0}^{r-b} a_{b,j} x^{2r-2b-2j}, \tag{2.6}$$

where an inner sum with negative upper limit is zero. Then

$$\sum_{n=0}^{\infty} \{W(x_n)A_2(n)A_{2b}(n) - R_{b,W}(x_n)\} = \sum_r w_r D_{b,r}. \tag{2.7}$$

The left side is an ordinary absolutely convergent series.

Proof. Subtract the first J terms in (2.4) from (2.1) and add back $\sum_{j<J} a_{b,j}\zeta(s+2b+2j, \frac{1}{2})$. The subtracted series converges normally in successively larger left half-planes as J increases. This gives the continuation and the list of possible poles. None is an even integer.

Apply the same subtraction to each monomial of W. After removing (2.6), the summand is $O(x_n^{-2})$. The added-back values are all $\zeta(-2\ell, \frac{1}{2})$, $\ell \geq 0$, and each is zero because $\zeta(s, \frac{1}{2}) = (2^s-1)\zeta(s)$. The evaluation of the meromorphic continuation is therefore exactly the displayed ordinary sum. The finite number of monomials permits termwise addition without an interchange issue. □

2.2 The inherited Bernoulli coordinate transform

Define transformed coordinates by

$$C_j = \sum_{r=0}^{j} \binom{2j+1}{2r} 2^{-2(j-r)} D_{b,r}. \tag{2.8}$$

The inverse transform is

$$D_{b,r} = \sum_{j=0}^{r} \frac{\binom{2r}{2j}}{2j+1} B_{2r-2j}(1/2) C_j. \tag{2.9}$$

Both identities follow from the generating function (1.4). Indeed the exact prefix polynomial is

$$\sum_{n=0}^{N-1} (n+\tfrac{1}{2})^{2r} = \sum_{j=0}^{r} \frac{\binom{2r}{2j}}{2j+1} B_{2r-2j}(1/2) N^{2j+1}. \tag{2.10}$$

The late coordinates C_j, $j \geq b$, close on ordinary even zeta values. The following is the precise inherited reduction, in the normalization used by the new inverse theorem [2, Sections 2–3].

Proposition 2.2 (Late-coordinate formula)**.** *Put $\mu_j = B_{2j}/(2j)!$ and $Z_k = \zeta(2k)$, including $Z_0 = -\frac{1}{2}$. For $j \geq b$,*

$$C_j = \sum_{k=0}^{b-1} \left\{ \frac{(2j-1)!}{(2b-2k)!} \mu_{j-b+k} + \mathbf{1}_{k=0} \frac{2j-2b+1}{2} B_{2j-2b} \right\} Z_k. \tag{2.11}$$

Consequently, if $\mathbf{C} = (C_b, \ldots, C_{2b-1})^{\mathsf{T}}$ and $\mathbf{Z} = (Z_0, \ldots, Z_{b-1})^{\mathsf{T}}$, then

$$\mathbf{C} = M_b \mathbf{Z}, \qquad (M_b)_{tk} = \frac{(2b+2t-1)!}{(2b-2k)!} \mu_{t+k} + \mathbf{1}_{k=0} \frac{2t+1}{2} B_{2t}. \tag{2.12}$$

We recall the finite argument to make the boundary convention explicit. For every odd integer $d > 0$, summation by parts uses the exact prefix expression

$$E_d(N) = N^d A_2(N) A_{2b}(N) + \sum_{k=1}^{N} k^{d-2b} A_2(k) + \sum_{k=1}^{N} k^{d-2} A_{2b}(k) + \sum_{k=1}^{N} k^{d-2b-2}. \tag{2.13}$$

It equals the moment sum whose prefix polynomial is N^d. For a nonnegative power m, the finite identity

$$\sum_{k=1}^{N} k^m A_p(k) = A_p(N) F_m(N) + \sum_{\ell=1}^{N} \frac{F_m(\ell-1)}{\ell^p}, \quad F_m(N) = \frac{B_{m+1}(N+1) - B_{m+1}(1)}{m+1}, \tag{2.14}$$

is obtained by reversing the finite sums, with the infinite part $A_p(N)$ left intact. Substituting it into (2.13), expanding the finite Bernoulli polynomials, and extracting the constant term gives, for $d = 2j+1 \geq 2b+1$, the two coefficients

$$B_{2j-2b+2k} \left\{ \frac{\binom{2j-2b+2}{2-2k}}{2j-2b+2} + \frac{\binom{2j}{2b-2k}}{2j} \right\}.$$

Here a binomial summand is omitted unless its lower index is at least two and at most its upper index. The first term consequently occurs only at $k = 0$. Cancelling the factorials gives (2.11). At the transition $j = b$, the two $-\gamma/2$ terms arising from the tail sums cancel the $+\gamma$ from the final harmonic sum in (2.13). This is why there is no extra Euler constant in the zeroth column. This finite computation is also the source proof of the proposition; it precedes the new reconstruction problem.

The binomial omission rule is essential. Adding a lower-index-zero term would change M_b. The matrix in (2.12), including its exceptional zeroth column, is the one inverted below.

2.3 Formal coordinates and the optimal prefix

The source regards

$$C_0, \ldots, C_{b-1}, \quad Z_0, \ldots, Z_{b-1} \tag{2.15}$$

as $2b$ formal labels. The Bernoulli transform from $D_{b,0}, \ldots, D_{b,2b-1}$ is triangular, with the late block M_b. Thus nonsingularity of M_b gives full rank of this prefix. It does not imply that the evaluated numbers in (2.15) are arithmetically independent. In fact ordinary even zeta values are rational multiples of powers of π. Our inverse theorem solves a completely specified rational linear problem on these formal labels and then evaluates the resulting identities analytically.

3 Every reconstruction row in explicit form

The main theorem of this section answers research question R2 of [2] for the complete $(2, 2b)$ family. The answer contains only factorials, Bernoulli numbers, and finite sums. In particular, it does not leave an inverse matrix or an unknown determinant in the definition of a reconstruction coefficient.

3.1 A finite Padé identity

For $N \geq 0$, define

$$P_N(t) = \sum_{k=0}^{N} \frac{(2N-k)!\, N!}{(2N)!\, k!\, (N-k)!} t^k. \tag{3.1}$$

The exact initial Taylor coefficients of $e^t P_N(-t) - P_N(t)$ vanish through degree $2N$, and

$$e^t P_N(-t) - P_N(t) = \frac{(-1)^N (N!)^2}{(2N)!(2N+1)!} t^{2N+1} + O(t^{2N+2}). \tag{3.2}$$

For completeness, the vanishing is the binomial identity

$$\sum_{k=0}^{N} (-1)^k \binom{N}{k} (1+x)^{2N-k} = x^N (1+x)^N$$

applied coefficient by coefficient. The first remaining coefficient is

$$\frac{1}{(2N)!} \sum_{k=0}^{N} \frac{(-1)^k \binom{N}{k}}{2N+1-k} = \frac{1}{(2N)!} \int_0^1 x^N (x-1)^N \, \mathrm{d}x,$$

which proves (3.2) by the beta integral.

Set

$$F(z) = \sum_{j \geq 0} \mu_j z^j = \frac{\sqrt{z}}{2} \coth \frac{\sqrt{z}}{2}. \tag{3.3}$$

The right side is an even germ in $\sqrt{z}$, hence a single-valued holomorphic germ in z. With $N = 2n+1$, put

$$q_n(t^2) = \frac{P_{2n+1}(t) - P_{2n+1}(-t)}{t}, \qquad r_n(t^2) = \frac{P_{2n+1}(t) + P_{2n+1}(-t)}{2}.$$

Then (3.2) implies

$$F(z) q_n(z) - r_n(z) = h_n z^{2n+1} + O(z^{2n+2}), \qquad h_n = \frac{((2n+1)!)^2}{(4n+2)!(4n+3)!}. \tag{3.4}$$

3.2 Two explicit orthogonal families

Let $\mathcal{L}$ be the rational linear functional on $\mathbb{Q}[x]$ with $\mathcal{L}(x^j) = \mu_j$. Define

$$\pi_n(x) = \sum_{t=0}^{n} p_{nt} x^t, \qquad p_{nt} = \frac{2(2n+2t+1)!(2n+1)!}{(4n+2)!(2n-2t+1)!(2t)!}, \tag{3.5}$$

$$c_n = \frac{(2n+1)(2n+2)!}{(4n+2)!}. \tag{3.6}$$

Thus $\pi_n(x) = x^n q_n(1/x)$ is monic. Coefficient comparison in (3.4) gives

$$\mathcal{L}(\pi_n) = c_n, \qquad \mathcal{L}(x^k \pi_n) = 0 \ (1 \leq k \leq n), \qquad \mathcal{L}(x^{n+1} \pi_n) = h_n. \tag{3.7}$$

These are the inherited shifted-functional polynomials. The new inverse construction requires orthogonality for $\mathcal{L}$ itself.

Lemma 3.1 (The unshifted Bernoulli family)**.** *Put*

$$Q_0(x) = 1, \qquad \nu_0 = 1,$$
$$Q_n(x) = \pi_n(x) - \frac{c_n}{c_{n-1}} \pi_{n-1}(x), \qquad \nu_n = -\frac{c_n}{c_{n-1}} h_{n-1} \quad (n \geq 1). \tag{3.8}$$

Then the Q_n are monic, pairwise orthogonal for $\mathcal{L}$, and $\mathcal{L}(Q_n^2) = \nu_n \neq 0$. Write $q_{nt} = [x^t] Q_n(x)$, with $q_{nt} = 0$ if $t > n$.

Proof. The zeroth moment of Q_n vanishes by (3.7). For $1 \le k < n$, both terms in $\mathcal{L}(x^k Q_n)$ vanish by the same formula. Therefore Q_n is orthogonal to every polynomial of degree less than n. As Q_n is monic,

$$\mathcal{L}(Q_n^2) = \mathcal{L}(x^n Q_n) = -\frac{c_n}{c_{n-1}}\mathcal{L}(x^n \pi_{n-1}) = -\frac{c_n}{c_{n-1}} h_{n-1}.$$

All factorials in c_n, h_n are positive, so the norm is nonzero. The functional is not positive: $\nu_0 > 0$ and $\nu_n < 0$ for $n \ge 1$. The proof is algebraic and uses no positivity assumption. □

Theorem 3.2 (Explicit inverse Bernoulli kernel)**.** *For $b \ge 1$, let $H_b = (\mu_{t+k})_{0\le t,k<b}$ and set*

$$K_b(x,y) = \sum_{n=0}^{b-1} \frac{Q_n(x)Q_n(y)}{\nu_n}, \qquad K_{kt}^{(b)} = [x^k y^t]K_b(x,y) = \sum_{n=0}^{b-1} \frac{q_{nk}q_{nt}}{\nu_n}. \tag{3.9}$$

Then H_b is nonsingular and $H_b^{-1} = (K_{kt}^{(b)})_{0\le k,t<b}$. Moreover,

$$K_{0t}^{(b)} = \frac{p_{b-1,t}}{c_{b-1}}. \tag{3.10}$$

Proof. Every polynomial f of degree less than b has the orthogonal expansion

$$f(x) = \sum_{n=0}^{b-1} \frac{\mathcal{L}(fQ_n)}{\nu_n} Q_n(x).$$

It follows that $\mathcal{L}_y(K_b(x,y)f(y)) = f(x)$. Taking $f(y) = y^j$ and comparing coefficients of x^k gives $\sum_t K_{kt}^{(b)} \mu_{t+j} = \delta_{kj}$, which proves the matrix identity. The polynomial π_{b-1}/c_{b-1} has moments one at degree zero and zero at degrees $1, \dots, b-1$ by (3.7); uniqueness of the first inverse row proves (3.10). □

3.3 The complete mixed inverse

Define the following finite rational data, for $0 \le t, k < b$:

$$u_t^{(b)} = \mu_t \frac{(2b)!(2t+1)!}{2(2b+2t-1)!}, \tag{3.11}$$

$$v_k^{(b)} = \sum_{t=0}^{b-1} K_{kt}^{(b)} u_t^{(b)}, \qquad \delta_b = 2 - B_{2b-2}. \tag{3.12}$$

Here every $K_{kt}^{(b)}$ is already given by the explicit finite sum (3.9) and the factorial coefficients (3.5)–(3.8).

Theorem 3.3 (All reconstruction rows)**.** *For every $b \ge 2$, $\delta_b \ne 0$, and*

$$\boxed{\mathcal{R}_{b;k,t} = \frac{(2b-2k)!}{(2b+2t-1)!}\left(K_{kt}^{(b)} - \frac{v_k^{(b)}}{\delta_b}\frac{p_{b-1,t}}{c_{b-1}}\right) = (M_b^{-1})_{kt}.} \tag{3.13}$$

Consequently every even-zeta coordinate is recovered by

$$\boxed{\zeta(2k) = \sum_{t=0}^{b-1} \mathcal{R}_{b;k,t} C_{b+t}, \qquad 0 \le k < b.} \tag{3.14}$$

This gives all rows in a common closed finite-sum formula.

Proof. Let

$$R_b = \operatorname{diag}_t (2b+2t-1)!, \qquad E_b = \operatorname{diag}_k \frac{1}{(2b-2k)!}.$$

The exact entries in (2.12) give

$$M_b = R_b (H_b + u^{(b)} e_0^{\mathsf{T}}) E_b. \tag{3.15}$$

By (3.10), factorial cancellation gives

$$v_0^{(b)} = \frac{1}{2b-1} \sum_{t=0}^{b-1} \binom{2b-1}{2t} (2t+1) B_{2t} = 1 - B_{2b-2}. \tag{3.16}$$

To prove the last equality directly, put $N = 2b-1 \geq 3$ and use

$$\sum_{j=0}^{N} \binom{N}{j} B_j = B_N(1) = 0, \qquad \sum_{j=0}^{N} j \binom{N}{j} B_j = -N B_{N-1}.$$

After removing the $j = 1$ terms and using the vanishing of the higher odd Bernoulli numbers, their sum is exactly (3.16).

We include the elementary nonvanishing argument. In the localization $\mathbb{Z}_{(2)}$ of rationals with odd denominator, the Bernoulli recurrence is

$$(2m+1)(2B_{2m}) = 2m - 1 - \sum_{j=1}^{m-1} \binom{2m+1}{2j} (2B_{2j}).$$

Induction and $\sum_{j=1}^{m-1} \binom{2m+1}{2j} = 2^{2m} - 2m - 2$ show that $2B_{2m}$ is an odd element of $\mathbb{Z}_{(2)}$. It cannot equal four. Hence $B_{2m} \neq 2$, proving $\delta_b \neq 0$.

Let $K = H_b^{-1}$ and $v = Ku$. A direct multiplication, using $1 + e_0^{\mathsf{T}} v = \delta_b$, shows

$$(H_b + u e_0^{\mathsf{T}})^{-1} = K - \frac{v e_0^{\mathsf{T}} K}{\delta_b}.$$

Insert (3.10) and multiply by the two inverse diagonal factors from (3.15). This is (3.13). Finally apply it to (2.12). □

Remark 3.4 (Recovery of the earlier rational row). The row $k = 0$ simplifies to

$$\mathcal{R}_{b;0,t} = \frac{2}{(2b-1)(2 - B_{2b-2})} \binom{2b-1}{2t}. \tag{3.17}$$

Thus the previously known rational extraction is recovered exactly. Rows $k = 1, \ldots, b-1$ in (3.13) are the missing rows asked for in the source. The first nontrivial complete inverse is

$$M_2 = \begin{pmatrix} 3/4 & 1/4 \\ 2/3 & -1/12 \end{pmatrix}, \qquad M_2^{-1} = \frac{1}{11} \begin{pmatrix} 4 & 12 \\ 32 & -36 \end{pmatrix}. \tag{3.18}$$

These small matrices are included as a transparent normalization check.

4 Evaluated ordinary series and polylogarithmic boundary identities

4.1 A polynomial for each prescribed even zeta value

For $0 \le k < b$, define the odd polynomial and its centered difference

$$S_{b,k}(y) = \sum_{t=0}^{b-1} \mathcal{R}_{b;k,t} y^{2b+2t+1}, \tag{4.1}$$

$$W_{b,k}(x) = S_{b,k}(x+\tfrac{1}{2}) - S_{b,k}(x-\tfrac{1}{2}), \qquad R_{b,k}(x) = R_{b,W_{b,k}}(x). \tag{4.2}$$

Here $R_{b,W}$ is the explicit asymptotic polynomial in (2.6). Thus $W_{b,k}$ is an even rational polynomial of degree at most $4b-2$, and $R_{b,k}$ is even of degree at most $2b-2$.

Theorem 4.1 (All even-zeta extraction series)**.** *For every $b \ge 2$ and every $0 \le k < b$,*

$$\boxed{\sum_{n=0}^{\infty} \Big\{ W_{b,k}(x_n)(\zeta(2) - H_n^{(2)})(\zeta(2b) - H_n^{(2b)}) - R_{b,k}(x_n) \Big\} = \zeta(2k).} \tag{4.3}$$

The summand is $O(n^{-2})$, and the sum is ordinary and absolutely convergent. Equivalently, its tail product may be replaced by

$$\frac{\psi'(n+1)\psi^{(2b-1)}(n+1)}{(2b-1)!}. \tag{4.4}$$

Every coefficient in (4.3) has the finite formula (3.13); the theorem is constructive at arbitrary order.

Proof. For an odd monomial,

$$(x+\tfrac{1}{2})^{2j+1} - (x-\tfrac{1}{2})^{2j+1} = \sum_{r=0}^{j} \binom{2j+1}{2r} 2^{-2(j-r)} x^{2r}.$$

Consequently the continued moment associated with $W_{b,k}$ is $\sum_t \mathcal{R}_{b;k,t} C_{b+t} = \zeta(2k)$ by (2.8) and theorem 3.3. Apply proposition 2.1 to turn that continued value into the ordinary sum in (4.3). The polygamma form follows from (1.2). □

Proposition 4.2 (Uniqueness on the optimal formal prefix)**.** *Among even polynomials of degree at most $4b-2$, $W_{b,k}$ is the unique one whose moment has formal coordinate vector Z_k in (2.15).*

Proof. The centered difference maps odd polynomials of degree at most $4b-1$ bijectively onto even polynomials of degree at most $4b-2$: in the monomial bases its matrix is triangular with nonzero diagonal. Write the preimage of a candidate polynomial as $\sum_{j=0}^{2b-1} s_j y^{2j+1}$. Its formal moment is $\sum_j s_j C_j$. The first b labels in (2.15) force $s_0 = \cdots = s_{b-1} = 0$. The remaining row must solve $sM_b = e_k^{\mathsf{T}}$, whose unique solution is the row (3.13). This is a formal uniqueness statement; accidental relations between evaluated constants are not excluded. □

4.2 A compact trigamma–third-polygamma evaluation

The row $k=1$ for $b=2$ gives a relatively short new member of the family. With $x_n = n + \frac{1}{2}$, put

$$P(x) = 4032x^6 + 2480x^4 - 524x^2 - 23.$$

Then

$$\boxed{\sum_{n=0}^{\infty}\Big\{64(63x_n^2+2)-3P(x_n)A_2(n)A_4(n)\Big\}=528\zeta(2)=88\pi^2.} \tag{4.5}$$

Both large terms in braces must be combined before summation. Each separate sum diverges. Their combination is absolutely convergent.

Using $t_n = 2n+1$ gives the equivalent integral-coefficient polygamma identity

$$\boxed{\begin{aligned}\sum_{n=0}^{\infty}\Big\{&32(63t_n^2+8)\\&-(63t_n^6+155t_n^4-131t_n^2-23)\psi'(n+1)\psi'''(n+1)\Big\}=176\pi^2.\end{aligned}} \tag{4.6}$$

This is proved by the exact row identity, not by recognizing a decimal value. Numerically, (4.5) evaluates to

$$868.525187295863558457435207989101299907605548\ldots,$$

as does $88\pi^2$.

For a direct coefficient check, the inverse row from (3.18) is $(32,-36)/11$ and

$$S_{2,1}(y)=\frac{32y^5-36y^7}{11}.$$

Its centered difference is $-3P(x)/528$. The nonnegative-power part of its product with the two tails is $-64(63x^2+2)/528$, giving (4.5) immediately. This short calculation also fixes the sign of the subtraction.

4.3 An evaluation at the next tail order

To illustrate recovery of a coordinate beyond $\zeta(2)$, let

$$\begin{aligned}P_3(t)&=2145t^{10}+40725t^8+162302t^6\\&\quad+146110t^4+22161t^2+61,\\Q_3(t)&=137280t^4+1874240t^2+1263872.\end{aligned}$$

The row $k=2$ for $b=3$ yields

$$\boxed{\sum_{n=0}^{\infty}\{Q_3(t_n)-5P_3(t_n)A_2(n)A_6(n)\}=19520\zeta(4).} \tag{4.7}$$

Once again the combined summand is $O(n^{-2})$. The product is $\psi'(n+1)\psi^{(5)}(n+1)/120$. The supplied coefficient tables include all rows for $2\le b\le 6$; the generation code implements the theorem for any $b\ge 2$. Those tables are examples of an all-order formula, not the evidence on which its all-order validity rests.

4.4 Finite polylogarithmic generators

Let $\theta = z\,d/dz$. For integers $p,q\ge 2$, the ordinary generating function of the tail product is

$$G_{p,q}(z)=\sum_{n=0}^{\infty}A_p(n)A_q(n)z^n=\frac{N_{p,q}(z)}{1-z},\qquad |z|<1, \tag{4.8}$$

where

$$N_{p,q}(z) = \zeta(p)\zeta(q) - \zeta(q)\,\mathrm{Li}_p(z) - \zeta(p)\,\mathrm{Li}_q(z) + \mathrm{Li}_{p,q}(z) + \mathrm{Li}_{q,p}(z) + \mathrm{Li}_{p+q}(z). \tag{4.9}$$

This is the tail-product bridge already present in [7]. A short direct proof is useful: multiply the series by $1-z$, use $A_p(n-1) = A_p(n) + n^{-p}$, and then use

$$\sum_{n\geq 1} \frac{z^n H_n^{(q)}}{n^p} = \mathrm{Li}_{p,q}(z) + \mathrm{Li}_{p+q}(z).$$

All series are absolutely convergent in the open disc, which justifies each rearrangement.

Theorem 4.3 (Evaluated polylogarithmic boundary operators)**.** *For every $b \geq 2$ and $0 \leq k < b$, define*

$$\Phi_{b,k}(z) = W_{b,k}(\theta + \tfrac{1}{2})\frac{N_{2,2b}(z)}{1-z} - R_{b,k}(\theta + \tfrac{1}{2})\frac{1}{1-z}. \tag{4.10}$$

This is a finite differential expression in ordinary and depth-two polylogarithms, holomorphic for $|z| < 1$. It extends continuously to the closed unit disc, and

$$\boxed{\Phi_{b,k}(1) = \zeta(2k).} \tag{4.11}$$

Proof. Termwise differentiation of normally convergent power series gives

$$\Phi_{b,k}(z) = \sum_{n=0}^{\infty} \{W_{b,k}(x_n)A_2(n)A_{2b}(n) - R_{b,k}(x_n)\} z^n.$$

The coefficients belong to ℓ^1 by theorem 4.1. The Weierstrass test gives uniform convergence on $|z| \leq 1$, and the boundary value is exactly (4.3). Individual differentiated terms in (4.10) can diverge at $z = 1$; the combined expression has the proved continuous extension. □

For example, (4.5) has the finite functional form

$$\lim_{z\uparrow 1} \left\{ 64(63(\theta + \tfrac{1}{2})^2 + 2)\frac{1}{1-z} - 3P(\theta + \tfrac{1}{2})\frac{N_{2,4}(z)}{1-z} \right\} = 88\pi^2. \tag{4.12}$$

The theorem evaluates a prescribed cancellation of boundary jets. It does not claim a reduction of arbitrary depth-two polylogarithms to depth one. The exact operator and the correct boundary constant are both part of the identity.

5 Order eight: an infinite family beyond the proved small orders

The incoming *Bernoulli, Tornheim, and Digamma Identities* report [2, general small-determinant reduction and research question R1] proves the optimal-prefix conjecture when the smaller even order is 2, 4, or 6, and reduces every fixed smaller order to a finite determinant. The following result establishes a substantial part of the first remaining order. Its proof uses an exact symbolic determinant certificate and a uniform Bernoulli congruence.

Theorem 5.1 (The order-eight prefix outside one residue class)**.** *Let $p = 8$ and $q = 2b > 8$. If*

$$b \not\equiv 1 \pmod{16}, \tag{5.1}$$

then the first q centered moments have full formal rank q on the inherited coordinate list $C_0, \dots, C_{b-1}, Z_0, \dots, Z_{b-1}$. The assertion concerns the coefficient matrix; it does not assert arithmetic independence of the evaluated coordinates.

For $b \geq 8$ satisfying (5.1), the normalized determinant ratio ρ_b defined below has the exact valuation

$$v_2(\rho_b) = v_2(b-1) - 4. \tag{5.2}$$

No all-b valuation formula is asserted.

5.1 The inherited reduction, with its normalization fixed

Here the centered moments are the regular values $D_{8,2b}(-2r)$, $0 \leq r < 2b$, where

$$D_{8,2b}(s) = \sum_{n \geq 0} A_8(n) A_{2b}(n) x_n^{-s} \quad (\Re s > -2b - 5)$$

is continued by the same midpoint subtraction as in Proposition 2.1. The coordinates C_j use (2.8) with $D_{b,r}$ replaced by $D_{8,2b}(-2r)$. The zeta coordinates are again $Z_k = \zeta(2k)$.

Set $\mu_r = B_{2r}/(2r)!$ for $r \geq 0$ and $\mu_r = 0$ for $r < 0$, and put

$$H_{4,b} = (\mu_{i+j-3})_{0 \leq i,j < b}, \quad R_{4,b} = \operatorname{diag}_{0 \leq i < b}(2b + 2i - 7)!, \quad E_b = \operatorname{diag}_{0 \leq j < b} \frac{1}{(2b - 2j)!}.$$

The source's b-square late-coordinate matrix is explicitly

$$(M_{4,b})_{ik} = B_{2i+2k-6} \left\{ \frac{\binom{2i+2}{8-2k}}{2i+2} + \frac{\binom{2b+2i-6}{2b-2k}}{2b+2i-6} \right\}, \qquad 0 \leq i, k < b. \tag{5.3}$$

An entry with negative Bernoulli index is zero. A binomial summand is omitted unless its lower index is at least 2 and at most its upper index. Thus every matrix in the determinant ratio is specified. Define

$$\rho_b = \frac{\det M_{4,b}}{\det R_{4,b} \det E_b \det H_{4,b}}. \tag{5.4}$$

The inherited shifted-Hankel theorem makes every factor in the denominator nonzero. The first-q determinant is $\det M_{4,b}/(4b-1)!!$, so it suffices to prove $\rho_b \neq 0$.

Here is a four-square formula whose size is independent of b. For $0 \leq i, \ell \leq 3$ set

$$n_i = b - 4 + i, \qquad N_{i\ell} = 2(n_i - \ell) + 1, \qquad R_\ell = 2b - 6 + 2\ell,$$
$$f_\ell(n) = \frac{1}{(2\ell+1)!} \prod_{r=0}^{2\ell-1} (2n - r)(2n + 3 + r), \qquad F_{i\ell} = f_\ell(n_i).$$

An empty product is 1. Define the polynomial weight

$$W_{i\ell}(z) = (z + 2\ell + 1)(2n_i + 2\ell + z + 1)^{\underline{2i}},$$

where the underline denotes a falling factorial, and its finite Newton coefficients

$$d_{i\ell h} = \frac{1}{h!} \sum_{j=0}^{h} (-1)^{h-j} \binom{h}{j} W_{i\ell}(N_{i\ell} - j). \tag{5.5}$$

Introduce seven independent symbols $U_1, \dots, U_7$. Set

$$\mathcal{A}_{i\ell} = 2F_{i\ell} + \frac{R_\ell!}{(2n_i+1)(2n_i+2)(2\ell+2)!N_{i\ell}!} \sum_{\substack{1\le h\le 2i+1\\ h \bmod 2}} d_{i\ell h} N_{i\ell}^{\underline{h}} U_{4-i+\ell+(h-1)/2}. \tag{5.6}$$

The ratio of factorials is interpreted as a rational function of b by cancellation of finite products; its index difference $R_\ell - N_{i\ell} = 1 + 4\ell - 2i$ is fixed.

For every integer $b \ge 8$ the inherited small-determinant reduction gives

$$\rho_b = \left.\frac{\det \mathcal{A}}{\det F}\right|_{U_j = 2B_{2b-2j}}. \tag{5.7}$$

To check the normalization, reverse the four source columns and divide row i by its nonzero Padé constant c_{n_i}. The unperturbed entries become $F_{i\ell}$. For the perturbation, apply (5.5) to the source weighted Bernoulli sum, using the Newton identity $W_{i\ell}(z) = \sum_h d_{i\ell h}(N_{i\ell} - z)^{\underline{h}}$ and

$$\sum_{j=0}^{N} \binom{N}{j} B_j (N-j)^{\underline{h}} = N^{\underline{h}} B_{N-h}(1).$$

When $b \ge 8$, $N_{i\ell} - h \ge 2$ for every retained odd h, and the even h terms have odd Bernoulli index at least 3 and vanish. The excluded $j = 1$ term contributes exactly another $F_{i\ell}$. This explains the coefficient $2F$ in (5.6) and the factor $U_j = 2B_{2b-2j}$; neither can be dropped. At smaller b the exceptional B_0 and $B_1(1)$ terms require the direct finite calculation recorded below.

5.2 The exact polynomial certificate

Put

$$\begin{aligned} D_8(b) = {} & 256b^8 - 3584b^7 + 18496b^6 - 40544b^5 + 25984b^4 \\ & + 18424b^3 - 5856b^2 - 8316b - 2025. \end{aligned} \tag{5.8}$$

This integer is odd at every integer b, since every nonconstant coefficient is even.

Lemma 5.2 (Finite integer certificate)**.** *The matrices* (5.6) *and* F *satisfy*

$$\begin{aligned} \det F = {} & \frac{256}{4725}(b-2)(b-1)(2b-5)(2b-3)(4b-11)(4b-9) \\ & \cdot (4b-7)^2(4b-5)(4b-3)D_8(b), \end{aligned} \tag{5.9}$$

$$\frac{\det \mathcal{A}}{\det F} = \frac{P(b; U_1, \dots, U_7)}{576(2b-5)D_8(b)}, \tag{5.10}$$

where $P \in \mathbb{Z}[b, U_1, \dots, U_7]$*. It has* 94 *monomials in the seven* U *variables, with integer polynomial coefficients in* b*. Under the indicated polynomial substitution, it obeys*

$$\boxed{P\big(b; 1 + 4(b-j) + 16(b-j)^2 \ (1 \le j \le 7)\big) \equiv 16b^2 + 44b + 4 \pmod{64}.} \tag{5.11}$$

The congruence is coefficientwise in $\mathbb{Z}[b]$*.*

Proof. This is a finite algebraic certificate. The accompanying `rank_eight_certificate.json` gives every coefficient of P. The independent verifier rebuilds (5.5)–(5.6), without reading a precomputed matrix. In each row it clears the least common denominator and computes the determinant as the sum of its 24 signed permutation products in $\mathbb{Z}[b, U_1, \dots, U_7]$. Cross multiplication verifies (5.9) and (5.10) by exact coefficient comparison. It then performs the seven displayed substitutions and checks that every coefficient of the difference in (5.11) is divisible by 64.

These instructions, together with the finite formulas defining the entries, specify a reproducible exact certificate. The calculation uses neither interpolation from values of b nor floating-point determinants. In particular, it certifies polynomial identities for all parameters, rather than inferring them from a sample. □

5.3 A uniform Bernoulli congruence

Let $\mathbb{Z}_{(2)}$ denote the rational numbers with odd denominator; congruences below are in this ring.

Lemma 5.3 (Doubled Bernoulli numbers modulo 64)**.** *For every integer $m \geq 4$,*

$$\boxed{2B_{2m} \equiv 1 + 4m + 16m^2 \pmod{64}.} \tag{5.12}$$

Proof. First, $2B_{2m} \in \mathbb{Z}_{(2)}$ is odd for every $m \geq 1$. This follows inductively from

$$(2m+1)(2B_{2m}) = 2m - 1 - \sum_{j=1}^{m-1} \binom{2m+1}{2j} (2B_{2j}).$$

Indeed the sum of the binomial coefficients on the right is even, and the denominator $2m+1$ is odd. Thus all even Bernoulli numbers have 2-adic valuation at least -1, a fact also proved in the incoming report.

Apply the Faulhaber formula to $S_{2m}(8) = \sum_{a=0}^{7} a^{2m}$. Its final term is $8B_{2m}$. For $m \geq 2$, every other nonzero term with an even Bernoulli index contains at least 8^3; its valuation is at least 8 after the Bernoulli denominator is included. The B_1 and B_0 terms have still greater valuation. Division by $2m+1$ has no effect because that integer is odd. Consequently

$$2B_{2m} \equiv \frac{1}{4} S_{2m}(8) \pmod{64}.$$

When $m \geq 4$, each even summand $a^{2m}/4$ vanishes modulo 64. For the four odd summands, expand the integer powers $9^m = (1+8)^m$, $25^m = (1+24)^m$, and $49^m = (1+48)^m$. After division by 4, the first coefficients are

$$\frac{1 + 9^m + 25^m + 49^m}{4} \equiv 1 + 20m + 736\binom{m}{2} \pmod{64}.$$

The coefficient of $\binom{m}{3}$ is 31232, divisible by 64; all subsequent terms have valuation at least 10. Finally, $736 \equiv 32 \pmod{64}$, and $20m + 32\binom{m}{2} = 4m + 16m^2$. This proves the lemma. □

5.4 Proof of the order-eight theorem

For $b \geq 11$ all seven indices $m = b - j$ in (5.7) are at least 4. Since P has integer coefficients, Lemma 5.3 may be substituted into Lemma 5.2, giving

$$P(b; 2B_{2b-2}, \dots, 2B_{2b-14}) \equiv 16b^2 + 44b + 4 \equiv 4(b-1)(4b+15) \pmod{64}. \tag{5.13}$$

The last factor is odd. If $b \not\equiv 1 \pmod{16}$, then $v_2(b-1) \le 3$, so the right side of (5.13) has valuation at most 5, and

$$v_2(P) = v_2(b-1) + 2.$$

The denominator in (5.10) has valuation 6: $576 = 2^6 \cdot 9$, while $2b-5$ and $D_8(b)$ are odd. This proves (5.2) and nonvanishing for all the stated $b \ge 11$.

The remaining six values $5 \le b \le 10$ are computed from the original b-square source matrix, with every omitted binomial summand treated exactly as prescribed. Their nonzero ratios are

b	ρ_b
5	309713/27640
6	263589503/10455480
7	−6698504467/244114290
8	1826240201143/4860702000
9	−334092867653747/85606130610
10	711292958938006619/13975082830800

At $b = 8, 9, 10$ their valuations are $-4, -1, -4$, agreeing with (5.2). The cases $b = 5, 6, 7$ are asserted only as nonzero, not as instances of that valuation formula. This completes the proof of Theorem 5.1.

5.5 Limits of the result and an informative failed extrapolation

The unevaluated class $b \equiv 1 \pmod{16}$ remains outside this theorem. Computed values in it do not prove its general nonvanishing. Moreover, an apparent extension of (5.2), suggested by low-order experiments during this work, is false. Independent exact evaluation with ordinary rational Bernoulli numbers gives

b	$P \bmod 65536$	$v_2(\rho_b)$	$v_2(b-1)-4$
65	52224	4	2
129	24320	2	3

The residues have valuations 10 and 8, respectively; the denominator still has valuation 6. These are counterexamples to a tempting extrapolation made in the present investigation, not source errata. They do not refute the optimal-prefix conjecture: both displayed determinants are nonzero. A further proof for the remaining residue class must therefore allow different valuation behavior rather than extend the formula beyond its proved scope.

6 All regular odd centered cubic trigamma moments

The canonical centered-square chapter evaluates all regular nonpositive even moments of $\psi'(n+1)^2$. Its cubic counterpart has the opposite parity and a slightly larger, but still fixed, coordinate space. We give an explicit formula for every regular negative odd value, ordinary absolutely convergent identities realizing these values, and a finite polylogarithmic expression for their exponential generating function.

The cubic series itself has classical antecedents. In particular, Furdui evaluates $\sum_{n\ge1} n\psi'(n)^3$ in [9, Theorem 1(b)]; Hoffman's sums-of-tails calculus provides a general multiple-zeta framework [10]. The contribution here is the entire centered polynomial-subtracted family, its finite Bernoulli formula, and its explicit generator. The lowest convergent evaluation is included as its normalization, without a historical-priority claim.

6.1 Regular values and ordinary convergent sums

Put

$$A_n = \psi'(n+1) = \zeta(2) - H_n^{(2)}, \qquad x_n = n + \tfrac{1}{2}, \qquad \mathcal{U}(s) = \sum_{n=0}^{\infty} A_n^3 x_n^{-s}. \tag{6.1}$$

The last series converges normally in $\Re s > -2$. Define rational numbers

$$g_k = \sum_{\substack{a+b+c=k \\ a,b,c\geq 0}} B_{2a}(1/2)B_{2b}(1/2)B_{2c}(1/2), \qquad k \geq 0. \tag{6.2}$$

Thus $g_0 = 1$, $g_1 = -1/4$, and $g_2 = 13/120$. The Bernoulli convention throughout is $B_1 = -1/2$.

Proposition 6.1 (Continuation and the correct parity)**.** *The function $\mathcal{U}$ has a meromorphic continuation to the complex plane, with simple poles at $s = -2-2k$, $k \geq 0$, and residues g_k. Every $s = -(2r+1)$, $r \geq 0$, is regular. Writing*

$$U_r = \mathcal{U}(-2r-1), \tag{6.3}$$

one has the ordinary convergent identity

$$U_r = \sum_{n=0}^{\infty} \left\{ x_n^{2r+1} A_n^3 - \sum_{k=0}^{r-1} g_k x_n^{2r-2-2k} \right\}. \tag{6.4}$$

The inner sum is empty for $r = 0$; every bracket is $O(x_n^{-2})$.

Proof. The centered Hurwitz expansion gives, to arbitrary finite order,

$$A_n \sim \sum_{j\geq 0} B_{2j}(1/2) x_n^{-2j-1}, \qquad A_n^3 \sim \sum_{k\geq 0} g_k x_n^{-2k-3}.$$

Subtracting finitely many terms and adding their Hurwitz-zeta sums continues $\mathcal{U}$ to successive left half-planes. The only possible poles are the stated simple poles, with the stated residues. For $j \geq 1$, $B_{2j}(1/2)$ has sign $(-1)^j$, so every summand in g_k has sign $(-1)^k$; thus $g_k \neq 0$. At $s = -2r-1$ the subtracted sums are $g_k \zeta(2k+2-2r, 1/2) = 0$ for $0 \leq k < r$. Indeed $\zeta(-2m, 1/2) = 0$ for all $m \geq 0$. This proves (6.4) and its convergence. □

6.2 A finite formula in four constant coordinates

Set

$$C_1 = 9\zeta(3) - \frac{17}{4}\zeta(4), \tag{6.5}$$

$$R_j = -\frac{3}{2(2j-1)} \sum_{h=0}^{j-2} (2h+1) \binom{2j-1}{2h+3} B_{2j-4-2h} B_{2h} + \frac{1}{4}\delta_{j,3}, \qquad j \geq 2, \tag{6.6}$$

$$C_j = 9B_{2j-2}\zeta(3) - \frac{1}{2}\delta_{j,2}\zeta(2) + R_j, \qquad j \geq 2. \tag{6.7}$$

Here δ is the Kronecker delta. The first few coordinates are

$$\begin{aligned} C_2 &= \frac{3}{2}\zeta(3) - \frac{1}{2}\zeta(2) - \frac{1}{2}, \\ C_3 &= -\frac{3}{10}\zeta(3) - \frac{2}{5}, \\ C_4 &= \frac{3}{14}\zeta(3) - \frac{5}{56}, \\ C_5 &= -\frac{3}{10}\zeta(3) + \frac{7}{45}. \end{aligned} \tag{6.8}$$

Theorem 6.2 (All centered cubic moments)**.** *For every integer $r \geq 0$,*

$$\boxed{U_r = \frac{1}{2r+2} \sum_{j=1}^{r+1} \binom{2r+2}{2j} B_{2r+2-2j}(1/2) C_j.} \tag{6.9}$$

In particular every ordinary sum in (6.4) belongs to

$$\mathbb{Q} + \mathbb{Q}\zeta(2) + \mathbb{Q}\zeta(3) + \mathbb{Q}\zeta(4).$$

No multiple-zeta coordinate of increasing weight is required as r increases.

For example,

$$U_0 = \frac{9}{2}\zeta(3) - \frac{17}{8}\zeta(4), \tag{6.10}$$

$$U_1 = -\frac{1}{8}\zeta(2) - \frac{3}{4}\zeta(3) + \frac{17}{32}\zeta(4) - \frac{1}{8}, \tag{6.11}$$

$$U_2 = \frac{5}{48}\zeta(2) + \frac{47}{160}\zeta(3) - \frac{119}{384}\zeta(4) + \frac{3}{80}, \tag{6.12}$$

$$U_3 = -\frac{49}{384}\zeta(2) - \frac{257}{1120}\zeta(3) + \frac{527}{1536}\zeta(4) - \frac{99}{4480}. \tag{6.13}$$

Thus (6.11) evaluates $\sum_{n\geq 0}(x_n^3\psi'(n+1)^3 - 1)$, and (6.12) evaluates $\sum_{n\geq 0}(x_n^5\psi'(n+1)^3 - x_n^2 + 1/4)$. These are convergent series; the displayed subtractions are part of the identities.

6.3 Proof by finite summation and explicit cancellation

Let CT_N denote the coefficient of $N^0(\log N)^0$ in a cutoff asymptotic expansion. We use it only after finite identities and standard asymptotic expansions. Every partial sum of a nonnegative even power of x_n has zero constant coefficient as a polynomial in N. Consequently (6.4) implies

$$U_r = \mathrm{CT}_N \sum_{n=0}^{N-1} x_n^{2r+1} A_n^3. \tag{6.14}$$

The centered power sum is the even polynomial

$$\sum_{n=0}^{k-1} x_n^{2r+1} = \frac{B_{2r+2}(k+1/2) - B_{2r+2}(1/2)}{2r+2} = \frac{1}{2r+2} \sum_{j=1}^{r+1} \binom{2r+2}{2j} B_{2r+2-2j}(1/2) k^{2j}. \tag{6.15}$$

Since $A_{k-1} = A_k + k^{-2}$, finite summation by parts reduces (6.9) to

$$C_j = \mathrm{CT}_N \left\{ 3\sum_{k=1}^{N} k^{2j-2} A_k^2 + 3\sum_{k=1}^{N} k^{2j-4} A_k + \sum_{k=1}^{N} k^{2j-6} + N^{2j} A_N^3 \right\}. \tag{6.16}$$

It remains to prove that this expression is (6.5)–(6.7).

The first two coordinates. The elementary tail rearrangements

$$\sum_{k=1}^{\infty} A_k^2 = 3\zeta(3) - \frac{5}{2}\zeta(4), \qquad \sum_{k=1}^{\infty} \frac{A_k}{k^2} = \zeta_{\mathrm{MZ}}(2,2) = \frac{3}{4}\zeta(4)$$

give (6.5). For the first equality, include $k = 0$ and partition the two tail indices to obtain $2\zeta_{\mathrm{MZ}}(2,1) + \zeta(3) = 3\zeta(3)$, then remove A_0^2.

For $j = 2$, write $U_m(N) = \sum_{k=1}^{N} k^m A_k$ and $V_m(N) = \sum_{k=1}^{N} k^m A_k^2$. The same finite summation identities give

$$\begin{aligned} \mathrm{CT}_N\, U_0(N) &= \gamma - \zeta(2) + 1, \\ \mathrm{CT}_N\, V_2(N) &= -\gamma + \frac{1}{2}\zeta(2) + \frac{1}{2}\zeta(3) - \frac{2}{3}, \\ \mathrm{CT}_N(N^4 A_N^3) &= -\frac{3}{2}. \end{aligned}$$

For clarity, the first line follows from $U_0(N) = H_N - H_N^{(2)} + NA_N$. The second follows by using $\sum_{k<\ell} k^2 = \ell^3/3 - \ell^2/2 + \ell/6$ in the quadratic summation identity below; the needed additional constants are $\mathrm{CT}_N\, U_1(N) = 1/4 - \gamma/2$ and $U_{-1}(\infty) = \zeta(3)$. Substitution in (6.16) yields the displayed C_2.

The general coordinate. Let $m = 2j - 2 \geq 4$ and put

$$F_m(N) = \sum_{k=1}^{N} k^m = \sum_{\substack{a=1 \\ a \text{ odd}}}^{m+1} f_a N^a + \frac{1}{2}N^m, \qquad f_a = \frac{1}{m+1}\binom{m+1}{a} B_{m+1-a}.$$

Then

$$F_m(k-1) = \sum_{\substack{a=1 \\ a \text{ odd}}}^{m+1} f_a k^a - \frac{1}{2}k^m.$$

Finite summation by parts gives the exact relation

$$V_m(N) = F_m(N)A_N^2 + 2\sum_{k=1}^{N} \frac{F_m(k-1)A_k}{k^2} + \sum_{k=1}^{N} \frac{F_m(k-1)}{k^4}. \tag{6.17}$$

Its term $-U_{m-2}(N)$ cancels exactly against the corresponding linear-tail term in (6.16). This cancellation occurs before constants or divergent harmonic terms are taken.

For every positive odd q, another finite summation gives

$$\mathrm{CT}_N\, U_q(N) = \frac{2-q}{4} B_{q-1} - \frac{1}{2}\delta_{q,1}\gamma. \tag{6.18}$$

Indeed, if $F_q(N) = \sum_{k=1}^{N} k^q$, then

$$U_q(N) = F_q(N)A_N + \sum_{k=1}^{N} \frac{F_q(k-1)}{k^2}.$$

Use the uncentered expansion $A_N \sim \sum_{a\geq 1} B_{a-1}N^{-a}$. Only the N^2 and N^q terms of F_q can contribute to the constant when $q \geq 3$, yielding $(2-q)B_{q-1}/4$. The case $q = 1$ follows directly from $F_1(N) = (N^2+N)/2$ and $(N - H_N)/2$.

In (6.17), the $a = 1$ terms contribute $9f_1\zeta(3) = 9B_m\zeta(3)$ to (6.16). The $a = 3$ harmonic constants cancel by (6.18); all remaining nonnegative-power sums have constant zero. For odd $a \geq 3$,

$$[N^{-a}]A_N^2 = -B_{a-3}.$$

Combining these boundary coefficients with (6.18) leaves the rational sum

$$\frac{3}{2}\sum_{\substack{a=3\\ a \text{ odd}}}^{m+1} f_a(2-a)B_{a-3},$$

apart from the even boundary coefficient. The latter is exactly

$$\frac{3}{2}[N^{-m}]A_N^2 + [N^{-m-2}]A_N^3 = \frac{1}{4}\delta_{m,4}. \tag{6.19}$$

To see this, all odd-index Bernoulli numbers except $B_1 = -1/2$ vanish. The terms containing exactly one B_1 cancel the ordinary even-even convolution; the only surviving possibility is the B_1^2/B_1^3 contribution at $m = 4$. Substitute $a = 2h + 3$ and $m = 2j - 2$ to obtain (6.6)–(6.7). This proves Theorem 6.2.

6.4 A fixed coordinate space and a rational cubic identity

Treat $\zeta(2), \zeta(3), \zeta(4), 1$ as formal coordinates. The coefficient matrix of U_0, U_1, U_2, U_3 in this order is

$$M = \begin{pmatrix} 0 & 9/2 & -17/8 & 0 \\ -1/8 & -3/4 & 17/32 & -1/8 \\ 5/48 & 47/160 & -119/384 & 3/80 \\ -49/384 & -257/1120 & 527/1536 & -99/4480 \end{pmatrix}, \qquad \det M = \frac{51}{81920}. \tag{6.20}$$

Thus these first four values generate the entire formal coordinate space of the cubic family, and every later value is an explicit rational combination of them. This is a coefficient-space statement; it asserts no unproved arithmetic independence of the four constants.

One particularly simple elimination is

$$11U_0 + 198U_1 + 512U_2 + 224U_3 = -\frac{21}{2}. \tag{6.21}$$

Consequently, with $P(x) = 11x + 198x^3 + 512x^5 + 224x^7$, one obtains the entirely rational ordinary sum

$$\boxed{\sum_{n=0}^{\infty}\left\{P(x_n)\psi'(n+1)^3 - 224x_n^4 - 456x_n^2 - \frac{1414}{15}\right\} = -\frac{21}{2}.} \tag{6.22}$$

Its summand is $O(x_n^{-2})$. Equation (6.22) follows from the four proved values and their prescribed subtractions, so no numerical relation search or independence assumption enters it.

Another elimination isolates the odd zeta coordinate:

$$37U_0 + 2136U_1 + 11344U_2 + 7168U_3 = 252\zeta(3). \tag{6.23}$$

With $P_3(x) = 37x + 2136x^3 + 11344x^5 + 7168x^7$, this becomes

$$\boxed{\sum_{n=0}^{\infty}\left\{P_3(x_n)\psi'(n+1)^3 - 7168x_n^4 - 9552x_n^2 - \frac{1148}{15}\right\} = 252\zeta(3).} \tag{6.24}$$

It is again an ordinary absolutely convergent sum. Equations (6.22) and (6.24) demonstrate that the fixed four-coordinate description supplies exact extraction identities, in addition to individual moment evaluations.

6.5 The cubic generator from the known square generator

Let

$$\mathcal{G}_2(t)=\sum_{r\ge 0}\mathcal{T}(-2r)\frac{t^{2r}}{(2r)!},\qquad \mathcal{G}_3(t)=\sum_{r\ge 0}U_r\frac{t^{2r+1}}{(2r+1)!},$$

where $\mathcal{T}(s)=\sum_{n\ge 0}A_n^2x_n^{-s}$ for $\Re s>-1$, continued meromorphically elsewhere, is the centered-square Dirichlet series of the canonical chapter. The inherited explicit expression is

$$\mathcal{G}_2(t)=\frac{t}{2\sinh(t/2)}\left[3\zeta(3)-\frac{1}{4t}\int_0^t(t-u)u^2\coth(u/2)\,du\right]. \tag{6.25}$$

This is recalled with attribution to [8], not presented as an additional new square-moment theorem.

Theorem 6.3 (An analytic cubic moment generator)**.** *The series $\mathcal{G}_3$ converges for $|t|<2\pi$. On that disk,*

$$\boxed{\mathcal{G}_3(t)=\frac{1}{2\sinh(t/2)}\int_0^t(t-u)\left\{3\cosh(u/2)\mathcal{G}_2(u)-\frac{17}{4}\zeta(4)-\frac{1}{4}\zeta(2)u^2+\frac{u^4}{96}\right\}du.} \tag{6.26}$$

The expression is holomorphic and odd; its apparent singularity at zero is removable. The integration path lies in the disk and is otherwise arbitrary.

Proof. Put $C(t)=\sum_{j\ge 1}C_jt^{2j}/(2j)!$. The Bernoulli transform (6.9) gives

$$\mathcal{G}_3(t)=\frac{C(t)}{2\sinh(t/2)}.$$

Write $E(t)=\frac{t}{2}\coth(t/2)$ and

$$F(t)=\frac{1}{t}\int_0^t(t-u)uE(u)\,du=\sum_{h\ge 0}\frac{(2h+1)B_{2h}}{(2h+3)!}t^{2h+2}.$$

Direct coefficient multiplication of (6.6) yields

$$\begin{aligned}C''(t)&=9\zeta(3)E(t)-\frac{3}{2}E(t)F(t)-\frac{17}{4}\zeta(4)-\frac{1}{4}\zeta(2)t^2+\frac{t^4}{96}\\&=3\cosh(t/2)\mathcal{G}_2(t)-\frac{17}{4}\zeta(4)-\frac{1}{4}\zeta(2)t^2+\frac{t^4}{96}.\end{aligned}$$

In the second line use (6.25) and $E(t)=\cosh(t/2)t/(2\sinh(t/2))$. All these series converge in $|t|<2\pi$ by the Bernoulli generating functions. The initial conditions are $C(0)=C'(0)=0$. Twice integrating proves (6.26) and the stated analyticity and parity. □

6.6 A finite depth-two polylogarithmic expression

For $0<z<1$ use the convention

$$\mathrm{Li}_{a,b}(z)=\sum_{m>n\ge 1}\frac{z^m}{m^an^b}.$$

This is the one-variable multiple polylogarithm with the outer index carrying the argument. In the following formula, every displayed depth-two first index is at least two.

Theorem 6.4 (Closed polylogarithmic generator)**.** *For real $0 < t < 2\pi$, set $z = e^{-t}$ and define*

$$\begin{aligned}\mathcal{C}(t) = &\left(9\zeta(4) + 6t\zeta(3) - \frac{t^4}{16}\right)\operatorname{Li}_2(z) + \left(12\zeta(3) - \frac{t^3}{2}\right)\operatorname{Li}_3(z)\\ &- 3t^2\operatorname{Li}_4(z) - 12t\operatorname{Li}_5(z) - \frac{45}{2}\operatorname{Li}_6(z) - \frac{3}{2}t^2\operatorname{Li}_{2,2}(z)\\ &- 6t\left(\operatorname{Li}_{3,2}(z) + \operatorname{Li}_{2,3}(z)\right) - 9\left(\operatorname{Li}_{4,2}(z) + \operatorname{Li}_{2,4}(z)\right) - 12\operatorname{Li}_{3,3}(z)\\ &+ \frac{15}{2}\zeta(6) - 6\zeta(3)^2 + 6t\zeta(5) + \frac{1}{2}t^3\zeta(3) + \frac{1}{8}t^2\zeta(4) - \frac{1}{48}t^4\zeta(2) - \frac{t^6}{1440}.\end{aligned} \tag{6.27}$$

Then

$$\mathcal{G}_3(t) = \frac{\mathcal{C}(t)}{2\sinh(t/2)}. \tag{6.28}$$

Equation (6.26) specifies its single-valued holomorphic odd continuation throughout $|t| < 2\pi$.

Proof. For ordinary polylogarithms, $\frac{d}{dt}\operatorname{Li}_a(e^{-t}) = -\operatorname{Li}_{a-1}(e^{-t})$. For the depth-two convention just stated,

$$\frac{d}{dt}\operatorname{Li}_{a,b}(e^{-t}) = -\operatorname{Li}_{a-1,b}(e^{-t}) \quad (a \geq 2), \qquad \frac{d}{dt}\operatorname{Li}_{1,b}(e^{-t}) = -\frac{\operatorname{Li}_b(e^{-t})}{e^t - 1}.$$

Together with $\coth(t/2) = (1 + e^{-t})/(1 - e^{-t})$, these give, by finite differentiation, that $\mathcal{C}''$ is the integrand in (6.26).

The integration constants may be checked by anchored primitives. If $F_a'(t) = -F_{a-1}(t)$ and the displayed endpoint values exist, then

$$\int_0^t (t-u)u^k F_0(u)\,du = \sum_{j=0}^{k}(j+1)\frac{k!}{(k-j)!}t^{k-j}F_{j+2}(t) - (k+1)!F_{k+2}(0) + tk!F_{k+1}(0). \tag{6.29}$$

Apply this with $F_a = \operatorname{Li}_{s+a}(e^{-t})$, or $F_a = \operatorname{Li}_{a,b}(e^{-t})$ for $k \geq 1$. The only potentially divergent endpoint is kept in the convergent combination

$$\operatorname{Li}_{1,4}(e^{-t}) - \zeta(4)\operatorname{Li}_1(e^{-t}) \longrightarrow -\zeta_{\mathrm{MZ}}(4,1) - \zeta(5) = \zeta(2)\zeta(3) - 3\zeta(5).$$

The elementary depth-two reductions

$$\begin{aligned}\zeta_{\mathrm{MZ}}(2,3) &= \tfrac{9}{2}\zeta(5) - 2\zeta(2)\zeta(3), & \zeta_{\mathrm{MZ}}(3,2) &= 3\zeta(2)\zeta(3) - \tfrac{11}{2}\zeta(5),\\ \zeta_{\mathrm{MZ}}(4,2) + \zeta_{\mathrm{MZ}}(2,4) &= \zeta(2)\zeta(4) - \zeta(6), & 2\zeta_{\mathrm{MZ}}(3,3) &= \zeta(3)^2 - \zeta(6)\end{aligned}$$

reduce the endpoint constants to those in (6.27). Thus $\mathcal{C}(0) = \mathcal{C}'(0) = 0$ in the limiting sense, and (6.26) proves the claim. No differentiation of an identity known only at integral zeta orders is used. □

6.7 Verification scope and immediate continuation questions

The supplied script independently reconstructs the cutoff constants using Faulhaber polynomials and the uncentered trigamma expansion; it does not insert (6.6) into that reconstruction. It agrees with the closed formula for $1 \leq j \leq 18$, checks the generator coefficient conversion over the same range, and verifies the second derivative of (6.27) as an exact symbolic zero. Independent direct tail sums plus centered Hurwitz-tail corrections check $0 \leq r \leq 8$. The largest displayed residual in

the higher cutoff calculation is below $5 \cdot 10^{-76}$ at 110-digit working precision. Those residuals are numerical diagnostics, not certified interval bounds or replacements for the finite-sum proof.

The new result suggests three concrete questions. First, determine whether all regular centered moments of $\psi'(n+1)^4$ admit a similarly explicit fixed coordinate space, and derive its generator with every subtraction and endpoint constant specified. Second, identify the analogous regular-parity moment spaces for mixed products $\prod_i \psi^{(m_i)}(n+1)$, distinguishing formal coefficient rank from arithmetic independence. Third, determine whether the particular depth-two combination in (6.27) reduces to a prescribed finite class of classical polylogarithms; a failed numerical search would not prove that it cannot. None of these questions is settled by the cubic calculation above.

7 The next two-particle resonance: overlap subtraction and exact constants

The determinantal harmonic-resonance report explicitly proposed the next two-particle resonance as the first test of a higher-pole subtraction [4, Section 9]. We resolve that target for arbitrary positive shape parameters, and by joint holomorphy for their right half-planes. The result consists of an explicit normally convergent subtraction, its complete spectral derivative calculus, and a finite-cutoff constant proved independently from an exact discrete Pearson identity. These statements are proved here; no claim of exhaustive literature novelty is made.

7.1 Definitions and a finite identity

Let $\Re a, \Re c > 0$ and put

$$w_n(u) = \frac{\Gamma(n+a)\Gamma(n+c)}{\Gamma(n+a+c+1+u)\Gamma(n+1)}, \tag{7.1}$$

$$S_N(u) = \sum_{0 \le m < n \le N} (n-m)^2 w_m(u) w_n(u), \qquad K_n(u) = \sum_{0 \le m < n} (n-m)^2 w_m(u) w_n(u). \tag{7.2}$$

Thus $S_N = \sum_{n=0}^{N} K_n$ and $K_0 = 0$. There is no factor 2!. Use the abbreviations

$$\mu = \frac{1}{ac}, \qquad h = \frac{1}{a} + \frac{1}{c}, \qquad q = 1 + h, \qquad s = ac + a + c, \tag{7.3}$$

$$\rho = \psi(1) - \psi(a+1) - \psi(c+1), \qquad \tau = \psi'(1) - \psi'(a+1) - \psi'(c+1), \qquad r_2 = \rho^2 + \tau. \tag{7.4}$$

The letter q in this section is a scalar abbreviation, not a cyclotomic level or an integration order.

Proposition 7.1 (Exact Pearson reduction)**.** *Write $M_j(N,u) = \sum_{n=0}^{N} n^j w_n(u)$ and $t_n(u) = n(n+a+c+u)w_n(u)$. For every $N \ge 0$,*

$$acM_0(N,u) - uM_1(N,u) = t_{N+1}(u), \tag{7.5}$$

$$(1-u)M_2(N,u) + (s+u)M_1(N,u) = Nt_{N+1}(u). \tag{7.6}$$

Consequently, at $u = 0$,

$$acM_0(N,0) = (N+1)(N+1+a+c)w_{N+1}(0), \tag{7.7}$$

$$S_N(0) = acNM_0(N,0)^2 - sM_0(N,0)M_1(N,0) - M_1(N,0)^2. \tag{7.8}$$

Proof. The exact weight ratio is

$$\frac{w_{n+1}(u)}{w_n(u)} = \frac{(n+a)(n+c)}{(n+a+c+1+u)(n+1)}.$$

It gives

$$t_{n+1} - t_n = (ac - un)w_n, \qquad (n+1)t_{n+1} - nt_n = ((1-u)n^2 + sn + ac)w_n.$$

Sum these identities, use $t_0 = 0$, and eliminate acM_0 to obtain (7.5)–(7.6). The elementary identity $S_N = M_0M_2 - M_1^2$ follows by expanding $(n-m)^2$ and pairing the two strict orderings. Substitution at zero proves (7.8). □

7.2 The Gamma product and its two different singular scales

Define holomorphic germs

$$R(u) = \frac{\Gamma(a+1)\Gamma(c+1)\Gamma(1+u)}{\Gamma(a+1+u)\Gamma(c+1+u)}, \qquad A(u) = \mu R(u), \tag{7.9}$$

$$\beta(u) = -s - (a+c+\tfrac{1}{2})u - \tfrac{1}{2}u^2, \qquad V(u) = \frac{1}{1-u^2}, \tag{7.10}$$

$$P(u) = (1 + hu + \mu u^2)R(u)^2, \qquad T(u;\alpha) = A(u)(\beta(u) + \alpha u). \tag{7.11}$$

In particular $R(0) = 1$, $R'(0) = \rho$, and $R''(0) = r_2$. For $\Re u > 1$, the partition converges absolutely, and Gauss summation [15, 15.4.20] gives

$$\mathcal{Z}(u) := \sum_{n\geq 0} K_n(u) = \frac{P(u)}{u^2(u-1)}. \tag{7.12}$$

For completeness, its three moment sums in this domain are

$$M_0(\infty, u) = \mu R(u), \qquad M_1(\infty, u) = \frac{R(u)}{u}, \qquad M_2(\infty, u) = \frac{R(u)(s+u)}{u(u-1)}.$$

Their determinant is (7.12). Thus the product needs only ordinary Gauss summation, not an assumed higher-dimensional Selberg formula.

Fix $\Re\alpha > 0$ and write $y = n + \alpha$, with the logarithm on the right half-plane. Define the *single grouped counterterm*

$$\begin{aligned} Q_n(u;\alpha) = {} & A(u)y^{-u} + T(u;\alpha)y^{-1-u} \\ & + \frac{2}{u}\left\{V(u)y^{-1-2u} - R(u)y^{-1-u}\right\}. \end{aligned} \tag{7.13}$$

The last line is holomorphic at $u = 0$ because its numerator vanishes there. The exponent $1 + 2u$ is essential: it accounts for the region in which the inner summation index also becomes large. Replacing it by $1 + u$ loses the logarithmic overlap term.

Theorem 7.2 (Normally convergent second-resonance germ)**.** *For $\Re a, \Re c, \Re\alpha > 0$, the series*

$$\mathcal{D}(u;\alpha) = \sum_{n\geq 0}\{K_n(u) - Q_n(u;\alpha)\} \tag{7.14}$$

converges absolutely and locally normally for u in a neighborhood of zero. It is jointly holomorphic in these parameters, and every fixed mixed parameter derivative can be taken termwise. Its exact value is

$$\begin{aligned} \mathcal{D}(u;\alpha) = {} & \frac{P(u)}{u^2(u-1)} - A(u)\zeta(u,\alpha) - T(u;\alpha)\zeta(1+u,\alpha) \\ & + \frac{2R(u)}{u}\zeta(1+u,\alpha) - \frac{2V(u)}{u}\zeta(1+2u,\alpha). \end{aligned} \tag{7.15}$$

All apparent poles at $u = 0$ cancel in this complete expression.

Proof. Work on compact shape-parameter sets and choose $0 < \delta < 1/4$. The locally uniform Gamma-ratio expansion [16, 5.11.13] is

$$w_n(u) = n^{-2-u}\left(1 + \frac{\beta(u)}{n} + O(n^{-2})\right), \qquad |u| \le \delta. \tag{7.16}$$

Gauss summation gives $\sum_{m\ge 0} w_m(u) = A(u)$ already in $\Re u > -1$. Hence, uniformly on the chosen compact sets,

$$\sum_{m<n} w_m(u) = A(u) - \frac{n^{-1-u}}{1+u} + O(n^{-2+\delta}). \tag{7.17}$$

For the first moment, the series $\sum_{m\ge 1}\{mw_m(u) - m^{-1-u}\}$ converges locally normally for $\Re u > -1$. On $\Re u > 0$ its value is $R(u)/u - \zeta(1+u)$, so that identity continues through zero. Its tail, together with the leading Hurwitz-zeta tail, gives the *holomorphic* estimate

$$\sum_{m<n} mw_m(u) = \frac{R(u) - n^{-u}}{u} + O(n^{-1+\delta}). \tag{7.18}$$

The quotient has its removable value at zero. Finally, summing $m^2 w_m(u) = m^{-u} + O(m^{-1+\delta})$ gives

$$\sum_{m<n} m^2 w_m(u) = \frac{n^{1-u}}{1-u} + O(n^{\delta}\log(n+1)). \tag{7.19}$$

Insert these estimates in

$$K_n = w_n\left(n^2 \sum_{m<n} w_m - 2n \sum_{m<n} mw_m + \sum_{m<n} m^2 w_m\right).$$

The coefficient of n^{-1-2u} is

$$-\frac{1}{1+u} + \frac{2}{u} + \frac{1}{1-u} = \frac{2}{u(1-u^2)}.$$

The resulting paired expression is holomorphic at zero, and

$$K_n = A(u)n^{-u} + A(u)\beta(u)n^{-1-u} + \frac{2}{u}\{V(u)n^{-1-2u} - R(u)n^{-1-u}\} + O(n^{-2+2\delta}\log(n+1)).$$

Changing n to $y = n + \alpha$ in the leading scale adds $\alpha u A(u) y^{-1-u}$; the remaining changes are summable uniformly. This proves (7.14), since the displayed error is summable. Cauchy's integral formula on slightly larger compact parameter sets proves the assertions for every fixed derivative.

To identify the sum, the same estimates hold on compact subsets of $-\delta < \Re u < 1 + \delta$, excluding $u = 1$. In particular, (7.19) still holds there: summing m^{-u} gives $n^{1-u}/(1-u)$ plus a locally bounded constant and $O(n^{\delta})$, while the partial-sum estimate for $m^2 w_m - m^{-u}$ is $O(n^{\delta}\log(n+1))$. Constants are uniform on compact sets avoiding $u = 1$. Thus the subtracted remainder remains summable throughout this strip. Both complete expressions are holomorphic on its connected punctured domain, including their removable values at zero. On $1 < \Re u < 1 + \delta$ all original sums and counterterms converge separately, and their difference is (7.15) by (7.12). The identity theorem gives the formula near zero. Its removability there was established by the normally convergent left side, independently of formal Laurent algebra. □

7.3 Exact hard-cutoff constant and spectral comparison

Set

$$\lambda = -q - 2\rho, \qquad C = -\rho^2 - q\rho - 2. \tag{7.20}$$

Theorem 7.3 (Closed constant at the next resonance)**.** *As $N \to \infty$,*

$$\boxed{S_N(0) = \mu N - \log^2 N + \lambda \log N + C + o(1).} \tag{7.21}$$

The error is $O(\log N/N)$ for fixed shape parameters, locally uniformly on their right half-planes. The spectral Laurent expansion is instead

$$\mathcal{Z}(u) = -\frac{1}{u^2} + \frac{\lambda}{u} + F + O(u), \qquad F = -\mu - q - 2q\rho - 2\rho^2 - \tau. \tag{7.22}$$

Thus the exact correction between the two constants is

$$F - C = -\mu - q + 2 - q\rho - \rho^2 - \tau. \tag{7.23}$$

Proof. At zero, (7.17)–(7.18) give

$$M_0(N,0) = \mu - \frac{1}{N} + O(N^{-2}), \qquad M_1(N,0) = \log N + \rho + O(N^{-1}).$$

Substituting them in the exact finite identity (7.8) gives

$$acNM_0^2 = \mu N - 2 + O(N^{-1}), \quad sM_0M_1 = q(\log N + \rho) + O(\log N/N),$$

and the asserted cutoff formula follows. This proof does not identify a spectral finite part with a cutoff constant. Separately expanding $R(u) = 1 + \rho u + (\rho^2 + \tau)u^2/2 + O(u^3)$ in (7.12) proves (7.22) and (7.23). □

With the Stieltjes convention

$$\zeta(1+u,\alpha) = \frac{1}{u} + \sum_{k\ge 0} \frac{(-1)^k}{k!}\gamma_k(\alpha)u^k, \qquad \gamma_0(\alpha) = -\psi(\alpha),$$

the constant term of Theorem 7.2 is the ordinary single-bracket identity

$$\boxed{\sum_{n\ge 0}\left\{K_n(0) - \mu - \frac{\lambda - 2\log(n+\alpha)}{n+\alpha}\right\} = C - \mu - \lambda\gamma_0(\alpha) + 2\gamma_1(\alpha).} \tag{7.24}$$

Its agreement with (7.21) also follows by summing the elementary counterterms before taking the limit. The coefficient $2\gamma_1$ comes from the doubled exponent $2u$.

7.4 Every derivative and an explicit first row

Write $R_j = R^{(j)}(0)$, $T_j = \partial_u^j T(0;\alpha)$ and $V_j = V^{(j)}(0)$, so $V_j = j!$ for even j and $V_j = 0$ for odd j. Let

$$\mathcal{N}(u) = \frac{P(u)}{u-1} + 2R(u) - V(u) - uT(u;\alpha). \tag{7.25}$$

Direct substitution gives $\mathcal{N}(0) = \mathcal{N}'(0) = 0$. Thus $\mathcal{N}(u)/u^2$ is an explicitly given holomorphic Gamma/polygamma germ, not an additional unspecified constant.

Corollary 7.4 (Finite formula for every spectral row)**.** *For every $k \ge 0$, put $D_k(\alpha) = \partial_u^k \mathcal{D}(0;\alpha)$. Then*

$$\begin{aligned} D_k(\alpha) = \frac{\mathcal{N}^{(k+2)}(0)}{(k+1)(k+2)} &- \mu \sum_{j=0}^{k} \binom{k}{j} R_j \zeta^{(k-j)}(0,\alpha) \\ &- \sum_{j=0}^{k} \binom{k}{j} T_j (-1)^{k-j} \gamma_{k-j}(\alpha) \\ &+ \frac{2}{k+1} \sum_{j=0}^{k+1} \binom{k+1}{j} (R_j - 2^{k+1-j} V_j)(-1)^{k+1-j} \gamma_{k+1-j}(\alpha). \end{aligned} \tag{7.26}$$

Here $\zeta^{(j)}$ denotes differentiation in the order variable. The normally convergent left side is $\sum_{n\ge 0}\{K_n^{(k)}(0) - Q_n^{(k)}(0;\alpha)\}$. If $L_n = \log(n+\alpha)$, every counterterm is the finite polynomial

$$\begin{aligned} Q_n^{(k)}(0;\alpha) = \mu \sum_{j=0}^{k} \binom{k}{j} R_j (-L_n)^{k-j} &+ \frac{1}{n+\alpha} \sum_{j=0}^{k} \binom{k}{j} T_j (-L_n)^{k-j} \\ &+ \frac{2}{(k+1)(n+\alpha)} \sum_{j=0}^{k+1} \binom{k+1}{j} \left\{ V_j (-2L_n)^{k+1-j} - R_j (-L_n)^{k+1-j} \right\}. \end{aligned} \tag{7.27}$$

Proof. Put $J(u;\alpha) = \zeta(1+u,\alpha) - 1/u$. Rearranging (7.15) gives the manifestly regular expression

$$\mathcal{D} = \frac{\mathcal{N}(u)}{u^2} - \mu R(u)\zeta(u,\alpha) - T(u;\alpha)J(u;\alpha) + \frac{2}{u}\{R(u)J(u;\alpha) - V(u)J(2u;\alpha)\}.$$

Apply the product rule. Division of a germ vanishing to order m converts its $(k+m)$th derivative to its kth derivative divided by $(k+1)\cdots(k+m)$; this gives the two denominators in (7.26). The same argument applied to (7.13) gives (7.27). Normal convergence was already proved in Theorem 7.2. □

All coefficients in this formula are finite polygamma polynomials. For example, the complete exponential Bell polynomials give

$$R_j = \mathrm{Bell}_j(\ell_1, \dots, \ell_j), \quad \ell_j = \psi^{(j-1)}(1) - \psi^{(j-1)}(a+1) - \psi^{(j-1)}(c+1).$$

The other coefficients follow by multiplication by the explicit polynomials and rational functions in (7.10)–(7.11). In fact the apparently required derivative R_{k+2} always cancels. To see this without coefficient cancellation, put $W(u) = (R(u)-1)/u$ and $d = \mu(\alpha - a - c - \frac{1}{2})$. Then

$$\frac{\mathcal{N}(u)}{u^2} = -W(u)^2 - V(u) + R(u)\left\{ \frac{(q+\mu u)W(u) + q + \mu}{u-1} - d + \frac{\mu}{2}u \right\}. \tag{7.28}$$

Its kth derivative uses only $R_0, \dots, R_{k+1}$. Thus the kth identity needs shape polygammas only through order k, while its Stieltjes constants run through order $k+1$. The summand itself has the equally explicit expression

$$K_n^{(k)}(0) = \sum_{m<n} (n-m)^2 w_m(0) w_n(0)\, \mathrm{Bell}_k(x_1(m,n), \dots, x_k(m,n)), \tag{7.29}$$

where

$$x_j(m,n) = -\psi^{(j-1)}(m+a+c+1) - \psi^{(j-1)}(n+a+c+1).$$

At positive integer parameters these are ordinary generalized harmonic numbers and zeta constants; at arbitrary shifts they are their polygamma continuations.

The first derivative can be printed without Bell notation. Define

$$B(\alpha) = r_2 + q\rho + q - 3 + \mu(\tfrac{1}{2} - \alpha), \tag{7.30}$$

$$C_1 = -\rho r_2 - q\rho^2 - \frac{q}{2} r_2 - (2\mu + q + 1)\rho - q - \frac{\mu}{2}. \tag{7.31}$$

Then

$$Q_n'(0;\alpha) = \mu(\rho - L_n) + \frac{3L_n^2 - \lambda L_n - B(\alpha)}{n+\alpha}, \tag{7.32}$$

$$D_1(\alpha) = C_1 - \mu\zeta'(0,\alpha) + B(\alpha)\gamma_0(\alpha) + \lambda\gamma_1(\alpha) - 3\gamma_2(\alpha). \tag{7.33}$$

Here $K_n'(0)$ is minus the defining double summand multiplied by $\psi(m+a+c+1)+\psi(n+a+c+1)$. The term $\zeta'(0,\alpha) = \log\Gamma(\alpha) - \frac{1}{2}\log(2\pi)$ is classical [13, 25.11.18]. Thus this row is an explicit ordinary convergent identity connecting a polygamma-weighted double sum, log-Gamma, and the first three generalized Stieltjes constants.

For comparison, the corresponding unsubtracted finite cutoff has the full first-derivative expansion

$$\begin{aligned} S_N'(0) = {} & -\mu N\log N + \mu(\rho+1)N + \log^3 N - \frac{\lambda}{2}\log^2 N \\ & - (r_2 + q\rho + q - 3 + \mu)\log N + C_1 + \mu\rho + o(1). \end{aligned} \tag{7.34}$$

Indeed take $\alpha = 1$ in (7.32), sum through N, use $\sum_{j=1}^{N+1}\log j = \log\Gamma(N+2)$ and the defining Stieltjes limits, and apply Stirling's formula. The constants $\gamma_0, \gamma_1, \gamma_2$ cancel against (7.33); the $\log(2\pi)$ terms cancel as well.

7.5 A uniform bridge from spectral germs to finite cutoffs

For any fixed finite derivative order, one can obtain the full cutoff expansion without making new interchanges. If $Y = N + 1 + \alpha$, the normally convergent tail in Theorem 7.2 gives

$$\begin{aligned} S_N(u) = {} & \mathcal{Z}(u) - A(u)\zeta(u,Y) - \left\{T(u;\alpha) - \frac{2R(u)}{u}\right\}\zeta(1+u,Y) \\ & - \frac{2V(u)}{u}\zeta(1+2u,Y) + O(N^{-1+2\delta}\log(N+1)). \end{aligned} \tag{7.35}$$

The complete expression has removable poles at zero. The error is locally uniform for $|u| \le \delta < 1/4$, after shrinking a surrounding disk if needed for derivatives. Thus every fixed Taylor coefficient may be extracted, with a remainder tending to zero. The standard large-Y Hurwitz expansion then generates every $N\log^j N$ term, every logarithmic counterterm, and its constant. The overlap exponent $2u$ controls the higher logarithmic powers.

7.6 Shift differentiation and an anchored primitive

The subtraction point is auxiliary; it does not move the original particle grid. Nevertheless (7.24) has the exact laws

$$D_0(\alpha+1) - D_0(\alpha) = \frac{\lambda - 2\log\alpha}{\alpha}, \tag{7.36}$$

$$\frac{d}{d\alpha}D_0(\alpha) = (\lambda+2)\zeta(2,\alpha) + 2\zeta'(2,\alpha). \tag{7.37}$$

These follow from the Hurwitz shift and derivative identities [13, 25.11.3 and 25.11.17]. For any anchor with $\Re\alpha_0 > 0$, a fixed-constant primitive is

$$\int_{\alpha_0}^{\alpha} D_0(t)\,dt = (C-\mu)(\alpha-\alpha_0) + \lambda \log\frac{\Gamma(\alpha)}{\Gamma(\alpha_0)} + \zeta''(0,\alpha) - \zeta''(0,\alpha_0). \tag{7.38}$$

The logarithm is the analytic log-Gamma on the right half-plane. To verify the last term, differentiate $\partial_\alpha\zeta(u,\alpha) = -u\zeta(1+u,\alpha)$ twice in u at zero; the result is $\partial_\alpha\zeta''(0,\alpha) = 2\gamma_1(\alpha)$. This also explains why an anchored primitive involves the second order derivative of Hurwitz zeta rather than an unspecified new constant.

7.7 Two normalization checks and a rational specialization

If every weight is multiplied by an arbitrary holomorphic $L(u)$ with $L(0) = 1$, then K_n, Q_n, $\mathcal{D}$, and $\mathcal{Z}$ are all multiplied by $L(u)^2$. Therefore the cutoff at $u = 0$ and its constant C are unchanged. Writing $l_1 = L'(0)$ and $l_2 = L''(0)$, the spectral finite part becomes

$$\widetilde{F} = F + 2l_1\lambda - l_1^2 - l_2, \tag{7.39}$$

whereas $\widetilde{D}_1 = D_1 + 2l_1 D_0$. These identities explicitly track both amplitude derivatives relevant to a double pole.

At $a = c = 1$ the weight is $w_n(0) = 1/((n+1)(n+2))$ and

$$\mu = 1, \quad \rho = \gamma - 2, \quad \lambda = 1 - 2\gamma, \quad C = \gamma - \gamma^2.$$

An exact largest-index formula, obtained from the finite moments, is

$$K_{v-1}(0) = 1 + \frac{1}{v+1} - \frac{H_v}{v} - \frac{H_v}{v+1}, \qquad v \geq 1. \tag{7.40}$$

Its first value is zero, as required by the empty inner sum. Thus

$$\sum_{0\leq m<n\leq N} \frac{(n-m)^2}{(m+1)(m+2)(n+1)(n+2)} = N - \log^2 N + (1-2\gamma)\log N + \gamma - \gamma^2 + o(1). \tag{7.41}$$

For the unnormalized weights the spectral finite part is $F = \zeta(2) - 2 + 2\gamma - 2\gamma^2$. With the different normalization $L(u) = \Gamma(1+u)$, the entire spectral product simplifies to

$$L(u)^2\mathcal{Z}(u) = \frac{1}{u^2(u-1)(1+u)^2},$$

whose finite part is -2. Both have exactly the same finite-cutoff constant $\gamma - \gamma^2$. This is a concrete second-order check of (7.39).

The specialization of (7.24) at $a = c = \alpha = 1$ yields

$$\sum_{v\geq 1}\left\{\frac{1}{v+1} - \frac{H_v}{v} - \frac{H_v}{v+1} - \frac{1-2\gamma-2\log v}{v}\right\} = \gamma^2 - 1 + 2\gamma_1.$$

This scalar specialization reduces to a classical harmonic–Stieltjes relation; its novelty is not asserted. The general-shape theorem, the uniform overlap subtraction, and the all-order cutoff comparison are the substantive continuation of the proposed resonance problem.

7.8 A functional polylogarithm identity and convergent endpoints

The rational specialization has an exact disk identity as well as a cutoff expansion. Define, for $|z| < 1$,

$$\mathcal{G}(z) = \sum_{0 \le m < n} \frac{(n-m)^2 z^{n+1}}{(m+1)(m+2)(n+1)(n+2)}.$$

The finite formula (7.40) gives

$$\boxed{\mathcal{G}(z) = \frac{z}{1-z} - 1 - \frac{\log(1-z)}{z} - \mathrm{Li}_2(z) - \frac{1}{2}\left(1 + \frac{1}{z}\right)\log^2(1-z).} \tag{7.42}$$

All apparent singularities at zero are removable. The first term is $z^2/12$, corresponding to the configuration $(m, n) = (0, 1)$. To prove the formula, sum the four terms of (7.40); the two harmonic transforms follow by integrating $\sum_{v \ge 1} H_v z^v = -\log(1-z)/(1-z)$ once, either with dz or dz/z. Thus this is a functional certificate before any special-value substitution.

At $z = 1/2$, the classical dilogarithm reflection identity [17, 25.12.6] gives

$$\sum_{0 \le m < n} \frac{(n-m)^2 2^{-(n+1)}}{(m+1)(m+2)(n+1)(n+2)} = 2\log 2 - \log^2 2 - \frac{\pi^2}{12}. \tag{7.43}$$

For $p \ge 2$, the ordinary convergent endpoint

$$J_p = \sum_{0 \le m < n} \frac{(n-m)^2}{(m+1)(m+2)(n+1)(n+2)(n+1)^p}$$

has a finite all-p reduction. To distinguish multiple zeta values from Hurwitz zeta, write $\zeta_{\mathrm{MZ}}(j, 1) = \sum_{v > m \ge 1} v^{-j} m^{-1}$. Set $E_j = \zeta_{\mathrm{MZ}}(j, 1) + \zeta(j+1)$ for $j \ge 2$; Euler's height-one formula reduces every E_j to products of at most two ordinary zeta values [4, Section 6]. Then

$$J_p = \zeta(p) - E_{p+1} + \sum_{j=2}^{p} (-1)^{p-j}(\zeta(j) - E_j) + (-1)^{p-1}(1 - \zeta(2)). \tag{7.44}$$

Indeed partial fractions give

$$\sum_{v \ge 1} \frac{1}{v^p(v+1)} = \sum_{j=2}^{p} (-1)^{p-j}\zeta(j) + (-1)^{p-1},$$
$$\sum_{v \ge 1} \frac{H_v}{v^p(v+1)} = \sum_{j=2}^{p} (-1)^{p-j} E_j + (-1)^{p-1}\zeta(2);$$

substitute them in (7.40). In particular,

$$J_2 = 3\zeta(2) - 2\zeta(3) - \tfrac{5}{4}\zeta(4) - 1, \tag{7.45}$$
$$J_3 = 1 - 2\zeta(2) + 4\zeta(3) - \tfrac{5}{4}\zeta(4) + \zeta(2)\zeta(3) - 3\zeta(5). \tag{7.46}$$

The primitive of order one has a simple explicitly subtracted endpoint:

$$\sum_{v \ge 1} \frac{K_{v-1}(0) - 1}{v} = 1 - \zeta(2) - 2\zeta(3). \tag{7.47}$$

Its convergence follows from $K_{v-1}(0) - 1 = O(\log v / v)$.

7.9 What this resolves, and what remains

The next $r = 2$ resonance requested in the earlier report is resolved: two moment divergences require three asymptotic counterterm scales, the overlap term is explicit, normal convergence is proved, and the cutoff/spectral constants are compared including normalization changes. The result does not establish a general higher-resonance theorem at arbitrary particle number. Natural next problems are to identify the escape strata at $r \geq 3$, formulate an inclusion–exclusion subtraction for simultaneous escape, and discover finite Pearson or contiguous relations that reduce the number of independent divergent moments. Another precise question is whether the order saving proved in (7.28) survives at higher particle number, where several nontrivial residue amplitudes replace the single $R(u)$.

8 All spectral coefficients from ordinary logarithmic Mellin moments

The incoming *Differential Polygamma and Lerch Jets* report [5, Section 9] asks for spatial functionals selecting higher coefficients of its residue Dirichlet germ, with the exact logarithmic normalization and endpoint constants. The construction below answers that question at every order. The Mittag–Leffler expansion and the scalar residue filter are inherited inputs; the shift-difference Mellin transform, its ordinary subtractions, and the higher-coefficient formulas are the continuation proved here. This is a claim of progress relative to the inspected material, not a historical priority claim for all specializations of Mellin calculus.

8.1 The data and the spatial normalization

Let P be a polynomial in finitely many variables $u_0, u_1, \dots$ and write

$$f_P(x) = P(\psi(x), \psi'(x), \dots), \qquad \phi_P(Y) = P(Y, 0, 0, \dots).$$

Let its principal part at $-n$ be

$$\sum_{j=1}^{M} \frac{p_j(n)}{(x+n)^j}, \qquad \Delta p_j(n) = p_j(n) - p_j(n-1), \qquad P_0(x) = \sum_{j=1}^{M} p_j(0) x^{-j}.$$

Zero coefficients may be included. These finite Laurent coefficients are the harmonic polynomials supplied by the incoming report. In particular, $p_j(n) = O((1+\log n)^d)$ and $\Delta p_j(n) = O((1+\log n)^d/n)$ for some fixed d. Define

$$F_P(x) = f_P(x) - f_P(x+1) - P_0(x), \tag{8.1}$$

$$c_P(L) = \phi_P'(L) + p_1(0) = \sum_{r=0}^{R} c_r L^r, \tag{8.2}$$

$$h_P(x) = F_P(x) + \mathbf{1}_{[1,\infty)}(x) \frac{c_P(\log x)}{x}. \tag{8.3}$$

The cutoff is exactly 1, the same scale as the finite-part integral on $[0,1]$. A value assigned to the step function at 1 has no effect.

Theorem 8.1 (Ordinary Mellin realization of every residue coefficient)**.** *The function F_P extends analytically through $x = 0$, and*

$$h_P(x) = O((1+\log x)^d/x^2) \quad (x \longrightarrow \infty).$$

Consequently

$$A_P(u) = \int_0^\infty x^{-u} h_P(x)\, dx \tag{8.4}$$

is holomorphic in $-1 < \operatorname{Re} u < 1$. *Its derivatives there are ordinary absolutely convergent integrals. Put*

$$\mathcal{B}_P(u) = \frac{\sin \pi u}{\pi} \left(A_P(u) - \sum_{r=0}^{R} \frac{c_r r!}{u^{r+1}} \right). \tag{8.5}$$

This is a meromorphic germ at zero. If

$$D_j(v) = \sum_{n \geq 1} \frac{\Delta p_j(n)}{n^v}, \qquad D_P = D_1,$$

then the exact meromorphic identity is

$$\boxed{\mathcal{B}_P(u) = D_P(u) + \sum_{j=2}^{M} \frac{(u)_{j-1}}{(j-1)!} D_j(u + j - 1).} \tag{8.6}$$

Every series in the sum over $j \geq 2$ *is normally convergent on* $\operatorname{Re} u > -1$. *In the original strip* $0 < \operatorname{Re} u < 1$ *one may equivalently write*

$$\mathcal{B}_P(u) = \frac{\sin \pi u}{\pi} \int_0^\infty x^{-u} F_P(x)\, dx. \tag{8.7}$$

Proof. Subtract the normalized Mittag–Leffler formula for $f_P(x+1)$ from that for $f_P(x)$. The constants cancel. Reindexing a finite cutoff and passing to its limit gives

$$F_P(x) = \sum_{n \geq 1} \sum_{j=1}^{M} \frac{\Delta p_j(n)}{(x+n)^j}. \tag{8.8}$$

Indeed the uncancelled last term is a principal part evaluated at $x + N + 1$, and tends to zero as $O((1 + \log N)^d / N)$. The new series is normally convergent on compact sets away from the negative integers. In particular it is analytic at zero. This also follows directly from $f_P - P_0$ and $f_P(x+1)$ being analytic there.

On the positive real axis, the usual differentiated digamma asymptotics give

$$f_P'(x) = \frac{\phi_P'(\log x)}{x} + O((1 + \log x)^d / x^2).$$

Every monomial containing a positive polygamma derivative contributes only to the error. Integrating this estimate from x to $x+1$, and using $P_0(x) = p_1(0)/x + O(x^{-2})$, proves

$$F_P(x) = -\frac{c_P(\log x)}{x} + O((1 + \log x)^d / x^2).$$

Thus the subtraction in (8.3) proves all assertions about A_P, including differentiation under the integral sign.

For $0 < \operatorname{Re} u < 1$ absolute convergence permits insertion of (8.8). The elementary beta integral is

$$\int_0^\infty \frac{x^{-u}}{(x+n)^j}\, dx = n^{1-u-j} \frac{\Gamma(1-u)\Gamma(j-1+u)}{\Gamma(j)}.$$

Its multiplication by $\sin(\pi u)/\pi$ is $n^{1-u-j}(u)_{j-1}/(j-1)!$, including $j=1$. The coefficient bound above makes the resulting sum absolutely convergent. Also

$$\int_1^\infty x^{-u-1}(\log x)^r\,dx = \frac{r!}{u^{r+1}}.$$

These two computations prove (8.6) and (8.7) in the strip. The $j \geq 2$ series converge locally normally for $\operatorname{Re} u > -1$; the scalar residue filter continues D_P meromorphically there. Uniqueness of continuation proves the germ identity. No integral at a divergent endpoint has been differentiated without its stated subtraction. □

8.2 An explicit coefficient formula at every order

For $m, r \geq 0$ define the finite polynomial

$$E_{m,r}(L) = [u^r]e^{-Lu}\prod_{k=1}^{m}(1+u/k). \tag{8.9}$$

Here an empty product is one. The first three are

$$E_{m,0} = 1, \qquad E_{m,1} = H_m - L, \qquad E_{m,2} = \frac{(H_m-L)^2 - H_m^{(2)}}{2}.$$

Thus these are finite harmonic polynomials, or equivalently Bell polynomials with cumulants $H_m - L$ and $(-1)^{k-1}(k-1)!H_m^{(k)}$ for $k \geq 2$.

Write $\mu_k(P) = \int_0^\infty h_P(x)(\log x)^k\,dx$. For $q \geq 0$ put

$$\begin{aligned} B_q(P) = &\sum_{\substack{k\geq 0\\ 2k+1\leq q}} \frac{(-1)^{k+q-2k-1}\pi^{2k}}{(2k+1)!(q-2k-1)!}\,\mu_{q-2k-1}(P) \\ &- \sum_{\substack{0\leq r\leq R\\ q+r\ \mathit{meven}}} c_r r!\frac{(-1)^{(q+r)/2}\pi^{q+r}}{(q+r+1)!}. \end{aligned} \tag{8.10}$$

The first sum is empty for $q=0$. It is often clearer computationally to take the coefficient directly in (8.5); formula (8.10) specifies every sign and factorial.

Corollary 8.2 (Higher coefficients and the complete Stieltjes term)**.** *For every integer $q \geq 1$,*

$$\boxed{[u^q]D_P(u) = B_q(P) - \sum_{j=2}^{M}\sum_{n\geq 1}\frac{\Delta p_j(n)}{(j-1)n^{j-1}}E_{j-2,q-1}(\log n).} \tag{8.11}$$

The double sum is finite in j and absolutely convergent in n. If the inherited scalar residue polynomial is

$$R_P(L) = -\frac{\phi_P(L+i\pi)-\phi_P(L-i\pi)}{2i\pi}, \qquad R_P'(L) = \sum_{r=0}^{S} a_r L^r,$$

put $\delta_n = \Delta p_1(n) - R_P'(\log n)/n$. For every $q \geq 0$ its entire ordinary-series counterpart is

$$\boxed{[u^q]D_P(u) = \frac{(-1)^q}{q!}\left(\sum_{r=0}^{S} a_r\gamma_{q+r} + \sum_{n\geq 1}\delta_n(\log n)^q\right).} \tag{8.12}$$

The second series converges absolutely. In particular, the same scalar projection controls all *Stieltjes corrections. If $\phi_P = 0$, every displayed γ_{q+r} term vanishes.*

Proof. Expanding A_P in (8.5) gives (8.10). For $j \geq 2$,

$$\frac{(u)_{j-1}}{(j-1)!} n^{-u} = \frac{u}{j-1} e^{-u \log n} \prod_{k=1}^{j-2} (1 + u/k).$$

Taking the coefficient of u^q in the normally convergent identity (8.6) proves (8.11). The logarithmic factor grows by only a fixed power, so the convergence assertion persists at every q.

The scalar filter gives $\delta_n = O((1 + \log n)^d / n^2)$ and

$$D_P(u) = \sum_r a_r (-1)^r \zeta^{(r)}(1 + u) + \sum_{n \geq 1} \delta_n n^{-u}.$$

With the convention $\zeta(1+u) = u^{-1} + \sum_{m \geq 0} (-1)^m \gamma_m u^m / m!$, the coefficient of u^q in $(-1)^r \zeta^{(r)}(1+u)$ is $(-1)^q \gamma_{q+r}/q!$. This proves (8.12). □

For $q = 1$, the last sum in (8.11) is exactly the inherited higher-pole correction

$$J_P = \sum_{j=2}^{M} \sum_{n \geq 1} \frac{\Delta p_j(n)}{(j-1) n^{j-1}}.$$

Hence $[u]\mathcal{B}_P = J_P + [u]D_P$, and the prior finite-part theorem is recovered with its original endpoint constant:

$$\mathrm{FP} \int_0^1 f_P(x)\, dx = C_P + [u]\mathcal{B}_P(u), \qquad C_P = \frac{1}{2\pi} \int_{-\pi}^{\pi} \phi_P(i\theta)\, d\theta.$$

This is a normalization check, not a new proof of that antecedent.

Corollary 8.3 (The first new logarithmic spatial functional)**.** *If P is mixed, meaning $\phi_P = 0$, then*

$$\boxed{[u^2]D_P(u) = -\int_0^\infty h_P(x) \log x\, dx + \frac{\pi^2 p_1(0)}{6} - \sum_{j=2}^{M} \sum_{n \geq 1} \frac{\Delta p_j(n)(H_{j-2} - \log n)}{(j-1) n^{j-1}}.} \tag{8.13}$$

All integrals and series are ordinary and absolutely convergent.

The local coefficient $p_1(0)$ is essential even in the mixed case. It is the correction from $\sin(\pi u)/(\pi u)$, and its coefficient is invisible at first order. Thus disappearance of the Stieltjes terms does not permit omission of this second-order endpoint term.

8.3 Concrete polygamma and nonlinear examples

For $P = u_1^2$ one has

$$p_4(n) = 1, \quad p_2(n) = 2(H_n^{(2)} + \zeta(2)), \quad p_1(n) = 4(H_n^{(3)} - \zeta(3)).$$

All omitted principal-part coefficients vanish. Therefore

$$\mathcal{B}_{u_1^2}(u) = (4 + 2u)\zeta(u + 3). \tag{8.14}$$

Let $Q(x) = \psi'(1 + x)$. An ordinary all-order generating formula is obtained from (8.5) using

$$h_{u_1^2}(x) = \begin{cases} 2\{Q(x) - \zeta(2) + 2\zeta(3)x\}/x^2, & 0 < x < 1, \\ 2\{Q(x) - \zeta(2)\}/x^2, & x > 1. \end{cases}$$

Its quadratic coefficient yields the explicit identity

$$\boxed{\begin{aligned}&\int_0^1 \frac{\psi'(1+x)-\zeta(2)+2\zeta(3)x}{x^2}\log x\,dx\\&\quad+\int_1^\infty \frac{\psi'(1+x)-\zeta(2)}{x^2}\log x\,dx\\&\qquad=-\zeta''(3)-\zeta'(3)-\frac{\pi^2\zeta(3)}{3}.\end{aligned}} \tag{8.15}$$

Primes on ζ denote derivatives in the spectral order. The linear polygamma example provides a particularly simple independent check of the endpoint term in (8.13).

For an example retaining a nonlinear spatial integrand, take $P = u_1^3$. Write $Z(s,t) = \sum_{n>m\geq 1} n^{-s}m^{-t}$ in its convergent region. Finite Laurent extraction gives

$$\begin{aligned}&p_6 = 1, && p_5 = 0, && p_4 = 3c_2, && p_3 = 6c_3,\\&p_2 = 9c_4 + 3c_2^2, && p_1 = 12c_5 + 12c_2c_3, && c_r = H_n^{(r)} + (-1)^r\zeta(r).\end{aligned}$$

Inserting their exact differences into (8.6) proves the finite depth-two identity

$$\boxed{\begin{aligned}\mathcal{B}_{u_1^3}(u) = {}&\frac{u^3+9u^2+32u+48}{2}\zeta(u+5)\\&+6(u+2)\{Z(u+3,2)+\zeta(2)\zeta(u+3)\}\\&+12\{Z(u+2,3)-\zeta(3)\zeta(u+2)\}.\end{aligned}} \tag{8.16}$$

All Z functions in a neighborhood of zero are ordinary absolutely convergent strict series. Arbitrary higher order derivatives may be taken there; no identity restricted to integer arguments is being differentiated.

For clarity, put $K = \zeta(5) + \zeta(2)\zeta(3)$ and

$$U(x) = \frac{3Q(x)}{x^4} + \frac{3Q(x)^2}{x^2} - \frac{3\zeta(2)}{x^4} + \frac{6\zeta(3)}{x^3} - \frac{9\zeta(4)+3\zeta(2)^2}{x^2}.$$

The ordinary function for (8.4) is

$$h_{u_1^3}(x) = U(x) + \frac{12K}{x}\mathbf{1}_{(0,1)}(x).$$

The displayed cancellations make it bounded at zero; for numerical work its convergent Taylor series avoids loss of digits there. With ∂_1 denoting a genuine first-order spectral derivative, the second coefficient of (8.16) is

$$\begin{aligned}B_2(u_1^3) = {}&12\zeta''(5)+16\zeta'(5)+\tfrac{9}{2}\zeta(5)+6(\partial_1^2+\partial_1)Z(3,2)\\&+6\zeta(2)\{\zeta''(3)+\zeta'(3)\}+6\partial_1^2 Z(2,3)-6\zeta(3)\zeta''(2).\end{aligned} \tag{8.17}$$

It follows that the following nonlinear logarithmic integral has a finite, convergent depth-two evaluation:

$$\boxed{\int_0^\infty h_{u_1^3}(x)\log x\,dx = -B_2(u_1^3) - 2\pi^2 K.} \tag{8.18}$$

This evaluates the logarithmic spatial functional in a specified spectral family. It makes no claim that the retained derivatives of Z are arithmetically irreducible.

8.4 What this settles and what remains

The higher-spectral-coefficient question is resolved for every differential polynomial: the spatial kernel, the scale, all endpoint terms, the higher-pole corrections, and every Stieltjes coefficient are explicit. The inherited convergent-word compiler applies to the coefficients in (8.11): it only acquires powers of $\log n$, hence higher derivatives in the existing spectral arguments. Thus these formulas introduce no additional kind of multiple-zeta coordinate. They do not settle relations between the resulting evaluated coordinates. A separate next problem is to extend the construction to independent shifts, where the pole lattices no longer align under the unit difference, and to characterize which linear combinations eliminate all higher-depth spectral derivatives.

9 Audit conclusions, integration changes, and further research

9.1 The precise change in mathematical status

The inspection covered the source statements used by this investigation, their nearby conjectures, and the six incoming reports at the pinned revision. It was not a line-by-line certification of the entire collective manuscript. No new counterexample or invalid proof in those inherited statements was established. Several tempting shortcuts in the present derivations were rejected or corrected before delivery. The appropriate repository changes are the following additions and status refinements.

Target in the inspected material	Proved advancement	Boundary retained
All reconstruction rows, Bernoulli report R2	Theorem 3.3 gives every entry of M_b^{-1}; Theorems 4.1 and 4.3 give its ordinary and polylogarithmic evaluations.	The inverse is for the $(2, 2b)$ matrix. Formal coordinate rank is not arithmetic independence.
Optimal-prefix conjecture beyond $p = 6$	Theorem 5.1 proves the $p = 8$, $q = 2b$ family for $b \geq 5$, $b \not\equiv 1 \pmod{16}$.	The missing residue class and general unequal orders $p \geq 10$ are not settled by this paper.
Centered cubic tails	Theorem 6.2 evaluates every regular odd centered cubic trigamma moment; Theorems 6.3 and 6.4 give its analytic and finite polylogarithmic generators.	The lowest cubic evaluation has classical antecedents. No general higher-power closure theorem is claimed.
Next two-particle resonance	Theorems 7.2 and 7.3 establish the grouped subtraction and exact cutoff constant. Corollary 7.4 gives every spectral row.	The result concerns two particles and the specified next resonance. Arbitrary particle number and arbitrary resonance remain open.
Higher spatial spectral functionals	Theorem 8.1 and Corollary 8.2 give ordinary logarithmic Mellin formulas at every coefficient order.	The statement uses the inherited unshifted differential-polynomial residue germ. General independent spatial shifts need a new analysis.
Short Gaussian S_6 and revised S_8	No status change.	The current frozen candidates remain conjectural. The older rejected S_8 vector is a different statement.

For integration, the full inverse belongs immediately after the rational row in the Bernoulli moment-rank development. The rank-eight theorem is an additional case of the existing optimal-prefix programme, and should be attached with its congruence qualification. The cubic section belongs after the canonical centered-square chapter. The resonance theorem continues the determinantal report; the Mellin theorem continues the differential-polygamma report. These are additive changes. The package's integration notes identify the source files and supplied artifacts.

9.2 Normalization issues that must survive editing

The omitted binomial term. The lower-index-zero summand in the source tail matrix is not present. Restoring it changes the matrix, the exceptional zeroth column, and the reconstruction coefficients. Formula (2.12) and the direct finite-cutoff verification in the package fix this convention explicitly.

The zeroth power in Faulhaber's formula. The uniformly correct expression for $\sum_{k=1}^{N} k^m$, $m \geq 0$, is (2.14), with $B_{m+1}(1)$ at the lower endpoint. Replacing it by $B_{m+1}(0)$ is valid for $m \geq 1$ but fails at $m = 0$. This scope error was found in an early draft of the present article and corrected. It is not reported as an error in an inherited theorem.

Exceptional cubic boundary terms. The $-\zeta(2)/2$ term at $j = 2$ and the rational $1/4$ term at $j = 3$ in (6.7) are required. The latter is the surviving B_1^2/B_1^3 boundary contribution in (6.19). Deleting low-order exceptions because a generic parity pattern looks simpler would change the convergent identities.

Cutoff constants and spectral finite parts. At the second resonance, C in (7.20) and F in (7.22) are different precisely normalized quantities. Equation (7.39) includes both derivatives of the amplitude needed at a double pole. The example $a = c = 1$ gives the same cutoff constant $\gamma - \gamma^2$ but different spectral finite parts under two normalizations. This is a proved comparison, not an assertion that the inspected report made the incorrect identification.

A second-order endpoint constant after scalar cancellation. When $\phi_P = 0$, the scalar-filter Stieltjes terms disappear, but $\pi^2 p_1(0)/6$ remains in (8.13). It comes from the sine factor in the Mellin transform. Its absence at first order provides no reason to discard it at second order.

An extrapolation that is actually false. The rank-eight determinant's valuation follows a simple law on the proved residue classes, but the unrestricted extrapolation fails at $b = 65$ and 129, as documented in the rank-eight section. These are counterexamples to a guess considered during this investigation, not to the source optimal-prefix conjecture. The determinants at those points are nonzero. The missing class must be addressed by a different argument or a finer congruence analysis.

9.3 Exact certificates and independent diagnostics

The accompanying scripts separate identities over $\mathbb{Q}$ from numerical comparisons. The reconstruction verifier checks both products with the original binomial matrix through $b = 15$, the shifted and unshifted orthogonality identities, the exceptional inverse row, and the printed examples. A second implementation reconstructs 35 late-coordinate rows directly from finite summation and the

uncentered Hurwitz expansion; it does not insert the proposed matrix formula. Exact coefficient tables contain all 20 extraction rows for $2 \le b \le 6$.

The rank-eight computation is a finite symbolic certificate in an integer polynomial ring, followed by a proved congruence for the Bernoulli inputs. Checking all coefficients of that polynomial identity is what makes it apply to an infinite family. A list of specialized nonzero determinants would not suffice. The full polynomial and a generator that reconstructs it from the four-by-four matrix are supplied.

The cubic verifier independently reconstructs 18 coordinates from finite cutoffs, checks the generating-function coefficients, and differentiates the finite polylogarithmic numerator twice with exact zero residual. It also checks the rational and $\zeta(3)$ extraction identities. The resonance verifier checks the finite Pearson relations and the Laurent cancellations algebraically, and evaluates defining finite weight sums with an independently generated asymptotic tail. The Mellin verifier checks 56 exact principal-part and coefficient identities, then compares spatial quadrature with spectral expressions.

Numerical diagnostics use 50, 80, or 110 decimal working digits, as recorded by each script. The six reconstruction comparisons have largest absolute residual below 3.4×10^{-59}; the cubic moment checks through $r = 8$ have largest higher-cutoff residual below 4.8×10^{-76}. The two nonzero-parameter resonance comparisons, including a complex parameter, agree within 1.4×10^{-30}. The nine Mellin comparisons have largest relative residual below 1.1×10^{-49}. These are independently organized diagnostics, not rigorous interval enclosures. Cancellation can exhaust working precision at large polynomial degree, so the reports retain the cutoff, expansion order, and successive comparison rather than a single unqualified count of "correct digits." The analytic proofs establish the infinite-family statements.

9.4 Further research questions

The following questions retain a concrete exact target. They prioritize identities and reductions; estimates are tools for justifying the manipulations, rather than separate objectives.

Q1. Finish rank eight on $b \equiv 1 \pmod{16}$. Determine an arithmetic invariant of the four-by-four perturbation that cannot vanish in this residue class. Higher modulus versions of the Faulhaber congruence are available, but the counterexamples to the simple valuation law show that its exponent cannot be predicted merely from $v_2(b-1)$. A useful result would prove nonvanishing for all remaining b, or identify a complete, exact valuation recursion with a termination argument. Checking more values alone does not settle the question.

Q2. Prove the optimal prefix for general unequal even orders. For $(p, q) = (2a, 2b)$ with $a \ge 5$, the inherited reduction leaves an a-square perturbation determinant. Seek a structural factorization, a congruence stable in a, or an algebraic interpretation of its nonvanishing. The Bernoulli functional is indefinite, so a positive moment-matrix argument cannot simply be imported. The result should identify the exceptional arithmetic factors, not leave their nonvanishing as an implicit assumption.

Q3. Construct all higher-p reconstruction rows. The kernel of Theorem 3.2 and a finite-rank inverse formula suggest a route for $(4, 2b)$ and $(6, 2b)$, whose rank is already known. The target is an explicit coefficient formula comparable to (3.13), with no unknown minors. Translate those coefficients into ordinary polynomial-subtracted polygamma sums with one prescribed even-zeta coordinate. The independent finite-cutoff method in the package can verify their normalization before an all-order proof is attempted.

Q4. Extend the reconstruction to shifted and character tails. Replace $A_p(n)$ by a Hurwitz tail or a Dirichlet-character tail. Determine the correct generalized Bernoulli kernel and state the exact finite correction when the midpoint identity $\zeta(-2j, 1/2) = 0$ is lost. One can first ask for a single rational shift and a common character, then for distinct shifts. A claimed ordinary evaluation must include the changed subtraction constants.

Q5. Classify higher powers of the centered trigamma tail. For $d \geq 4$, examine $\sum_{n\geq 0} \psi'(n+1)^d x_n^{-s}$ at its regular integer values. Does the correct parity family lie in a fixed finite coordinate space for each d? The cubic proof identifies a finite cancellation of linear tails before taking constant terms. Find its higher-power analogue, or prove a specific obstruction. An explicit quartic formula and a generator would be a decisive next case; no quartic closure is assumed here.

Q6. Determine the functional depth of the cubic generator. Theorem 6.4 uses six specified depth-two polylogarithms. Determine whether their particular combination admits a smaller function basis under an explicitly stated class of argument transformations. An equality of derivatives together with an anchored value is a suitable certificate. A numerical relation at one e^{-t} does not establish a functional reduction, and formal symbol rank alone does not establish arithmetic irreducibility at an endpoint.

Q7. Integrate the new boundary operators at interior arguments. The operators in (4.10) prescribe exact cancellations at $z = 1$. Find anchored primitives that lower their differential order and give compact formulas at interior algebraic points, including $1/2$ and cyclotomic points accessible by continuation. Track all integration constants from the power series at zero. The rational resonance generator in (7.42) supplies a simple model of this approach.

Q8. Develop a general simultaneous-escape subtraction. For three or more particles, classify the regions in which subsets of summation indices tend to infinity together. Determine an exact inclusion–exclusion subtraction and the associated exponents in the Hurwitz terms. The doubled scale $1+2u$ in the present two-particle case is the first nontrivial example. A successful theorem must prove normal convergence of the full grouped expression before differentiating at a resonance.

Q9. Explain the saving of one shape-polygamma order. Equation (7.28) removes the apparent R_{k+2} dependence from every kth spectral row. Is this a consequence of a general determinant identity or of the two-particle Pearson reduction? Find a corresponding regular expression at higher resonances or prove where the saving fails. Such a result would reduce the number of independent harmonic weights needed in exact derivative formulas.

Q10. Extend the ordinary Mellin bridge to independent shifts. For P in polygamma functions evaluated at several distinct shifts, formulate the exact shift-difference principal parts and identify the replacement for $D_j(u + j - 1)$. A natural target is a normally convergent Lerch or Hurwitz multiple-zeta family with all endpoint logarithms explicit. The first problem is to specify the spatial subtraction scale and its transformation law; the generalized Stieltjes corrections then have an unambiguous meaning.

Q11. Reduce the concrete nonlinear spectral derivatives. The cube identity (8.16) provides particular combinations of $\partial_1^j Z(s, t)$ at convergent integer pairs. Determine which combinations reduce to ordinary zeta derivatives, height-one coefficients, or the existing Tornheim coordinates. Functional relations must hold in a neighborhood of the differentiated point; an identity true

only at integer orders cannot be differentiated as if it were such a relation. The logarithmic integral (8.18) is a concrete initial target.

Q12. Compute the spectral image of the differential quotient. The incoming report classifies integration by parts in a free differential-polynomial algebra. The Mellin bridge supplies a map from that quotient, with its endpoint functional, to harmonic residue series. Determine the kernel of this map at fixed weight, first exactly at low weight and then by structural generators. A formal quotient dimension and a dimension after substituting the digamma function need not agree.

Q13. Build anchored multivariable primitive systems. Combine the spectral derivatives in (7.26) with the shift primitive (7.38). Determine finite recurrences for iterated primitives in the shift and shape parameters, with base values fixed at specified anchors. The goal is an identity in known Gamma, Hurwitz-zeta, Stieltjes, and multiple-zeta families, rather than naming an undetermined constant after each integration.

Q14. Return to the frozen Gaussian S_6 and revised S_8 targets. Preserve their exact current vectors and the repository's established shuffle equivalences. Seek a functional or iterated-integral relation whose rigorously evaluated boundary is precisely one of those vectors. The present identities expand the available analytic tools but establish no direct implication for those two conjectures. Their status should change only after such a relation and its endpoint evaluation are proved.

The concrete deliverable of this investigation is therefore a set of proved reconstruction, moment, resonance, and Mellin theorems with explicit coefficients and normalization constants. The unresolved questions are kept attached to the exact missing mathematical step.

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